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Ton Duc Thang University

Faculty of Information Technology



Bui Phuong Nam – 522H0002

Hoang Minh Quan – 523002451

A Comparison of Monolithic, Modular Monolith, and Microservices Architectures in LMS Systems

**INFORMATION TECHNOLOGY PROJECTS
SOFTWARE ENGINEERING**

Supervisor

Msc. Mai Van Manh

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Certification

This Projects was carried out at **Ton Duc Thang University**.

Advisor: MSc. Mai Van Manh

.....
(Title, Full Name and Signature)

This project was defended at the Undergraduate Thesis Examination Committee held at Ton Duc Thang University on / /

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Declaration

This project was completed at **Ton Duc Thang University**. We hereby declare that this is our own research work, conducted under the academic supervision of **MSc. Mai Van Manh**. All analyses, discussions, and results presented in this project are truthful, original, and have not been published in any form prior to this submission.

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Abstract

The selection of an appropriate software architecture plays a critical role in the development and evolution of **Learning Management Systems (LMS)**, directly affecting system scalability, maintainability, and adaptability to changing requirements. This project presents a comparative study of three widely adopted architectural approaches—**Monolithic**, **Modular Monolith**, and **Microservices**—within the context of an LMS application.

A core LMS system was designed with essential functionalities, including user management, course management, enrollment, learning content delivery, and assessment handling. Based on the same functional requirements, the system was implemented separately using the three architectural styles to ensure a fair and consistent comparison. Several change and extension scenarios, such as modifying grading logic, adding learning progress statistics, and preparing for future learning analytics integration, were defined and applied to each architecture.

The architectures were evaluated using technical criteria including module isolation, change impact scope, scalability at the functional level, deployment and operational complexity, and suitability for different system scales. The experimental results highlight the strengths and limitations of each approach: Monolithic architecture offers simplicity and ease of deployment but exhibits limited flexibility when evolving; Modular Monolith improves maintainability through clearer module boundaries while retaining centralized deployment; Microservices provide superior scalability and change isolation at the cost of increased implementation and operational complexity.

This study provides practical insights into the real-world trade-offs of architectural choices and proposes recommendations for selecting suitable architectures for LMS systems of small, medium, and large scales.

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Chapter 1

Introduction

1.1 Background and Rationale

In recent years, the rapid growth of technology and the push for digital transformation in education have made the development of robust Learning Management Systems (LMS) more urgent than ever. The architecture behind such systems is not merely a technical choice; it directly shapes their scalability, maintainability, and long-term. Approaches such as Monolithic, Modular Monolith, and Microservices have emerged as dominant paradigms, each offering unique strengths and trade-offs. This essay will explore and contrast these three architectural models in the context of building an LMS, focusing on functional modules such as user authentication, course management, content delivery, assessment, and progress tracking.

1.2 Project Objectives

In general, the objectives of this project is building and deploying a Learning Management System (LMS) in form of three different software architectures.

For specific, the following targets will be expected:

1. Design an LMS system with core functions (CRUD, authentication / authorization, enrollment, learning content, assignment & grade, learning process,...)
2. Deploy the LMS system according to three different architectures
3. Develop scenarios for system change and expansion
4. Compare architectures based on specific technical criteria
5. Analyze the advantages and disadvantages of each architecture in actual implementation

1.3 Scope of the Project

The scope of the project focuses on researching, designing and comparing three software architectures when applied to the same business problem. The LMS system is only used as a research object to illustrate and verify the above architectures, not the main development goal of the project. In other words, the focus of the topic is software architecture analysis, while LMS construction only serves as a means for deployment and evaluation.

Across all three architectures, the system is consistently decomposed into core functional modules (e.g., authentication/authorization, enrollment, learning content, assignments and grading, progress tracking). Extended features such as learning analytics or recommendation algorithms are acknowledged but reserved for future work.

The project does not prioritize detailed user interface or user experience (UI/UX) design, employing only minimal interfaces sufficient for testing and demonstration. Regarding technology choices, the team adopts contemporary frameworks considered most suitable at present, namely NestJS and Vue.js due to their widespread adoption, strong community support, and suitability for modular.

Other architectures (e.g., serverless, large-scale event-driven architecture, or complex hybrid architectures) are outside the scope of the study.

1.4 Research Methodology

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1.4.1 Theoretical Methodology

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1.4.2 Practical Methodology

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1.5 Structure of the Report

This report is structured as follows:

- **Chapter 1:** Introduction - Provides background, objectives, scope, methodology, and references.
- **Chapter 2:** Literature Review - Reviews existing technologies and related work.
- **Chapter 3:** System Design and Implementation - Details the design and implementation of the messaging system.
- **Chapter 4:** Testing and Evaluation - Presents the testing procedures and evaluation results.
- **Chapter 5:** Conclusion and Future Work - Summarizes findings and suggests future directions.
- **Appendices:** Contains supplementary material and code listings.

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Chapter 2

Literature Review

2.1 Related Works

The field of real-time messaging has seen significant advancements in both academic research and industry practice. Early systems focused on basic message delivery, but modern solutions emphasize scalability, low latency, and reliability [?, ?, ?]. Comparative studies have evaluated the strengths and weaknesses of various protocols and architectures, providing valuable insights for system designers [?, ?].

2.1.1 Existing Real-Time Messaging Solutions

Numerous real-time messaging solutions have been developed, each with unique trade-offs in terms of scalability, protocol support, and developer experience [?, ?, ?]. For example, Socket.IO and WebSocket-based frameworks are widely adopted for their ease of use and performance, while cloud-hosted services like Firebase and Pusher offer managed infrastructure [?, ?].

Here are some popular real-time messaging solutions:

- **Socket.IO:** A widely used library for real-time web applications.
- **Firebase Realtime Database:** A cloud-hosted NoSQL database that supports real-time data synchronization.
- **Pusher:** A hosted service that provides real-time APIs for web and mobile applications.
- **SignalR:** A Microsoft library for adding real-time web functionality to applications.
- **MQTT:** A lightweight messaging protocol for small sensors and mobile devices, optimized for high-latency or unreliable networks.

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2.1.2 Research Studies on Real-Time Systems

Research on real-time systems has explored a variety of topics, including protocol efficiency, system reliability, and performance optimization [?, ?, ?]. Studies have shown that cloud-native deployment patterns and microservices architectures can significantly improve scalability and maintainability [?, ?].

2.2 Relevant Technologies

The development of real-time messaging systems leverages a broad set of technologies, from programming languages and frameworks to cloud platforms and monitoring tools [?, ?, ?]. Node.js, WebSocket, and Redis are particularly prominent in recent implementations [?, ?].

2.2.1 Node.js

Node.js is a powerful JavaScript runtime built on Chrome's V8 engine, designed for building scalable network applications. Its core architecture is event-driven and asynchronous, allowing it to handle thousands of concurrent connections efficiently. The non-blocking I/O model means that operations such as reading files or querying databases do not halt the execution of other code, which is crucial for real-time systems. This design enables Node.js to deliver high throughput and low latency, making it ideal for chat applications, streaming services, and collaborative tools. The event loop, a central component of Node.js, manages all asynchronous operations and ensures the system remains responsive under heavy load.

Node.js Architecture

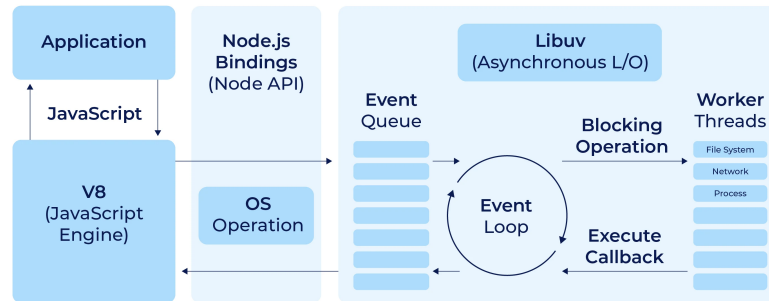


Figure 2.1: Node.js Architecture: Emphasizing Async, Event-Driven, Non-Blocking I/O

Node.js also benefits from a vast ecosystem of libraries and frameworks available through npm, which accelerates development and fosters innovation. Its single-threaded nature, combined with the event-driven paradigm, allows developers to write simple code that scales well across distributed systems. However, this architecture also requires careful handling of CPU-intensive tasks, as they can block the event loop and degrade performance. To address this, Node.js provides mechanisms such as worker threads and clustering to distribute workloads. Overall, Node.js has become a cornerstone technology for modern, real-time web applications due to its efficiency and flexibility.

2.2.2 WebSocket Protocol

The WebSocket protocol is a communication standard that enables persistent, full-duplex connections between clients and servers. Unlike traditional HTTP, which requires a new connection for each request-response cycle, WebSocket establishes a single, long-lived connection that remains open for continuous data exchange. This approach significantly reduces the overhead associated with connection setup and teardown, resulting in lower latency and improved performance for real-time applications. WebSocket is particularly well-suited for scenarios where instant updates are required, such as online gaming, financial trading platforms, and collaborative editing tools. Its ability to push data from the server to the client without repeated polling is a key advantage over HTTP. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

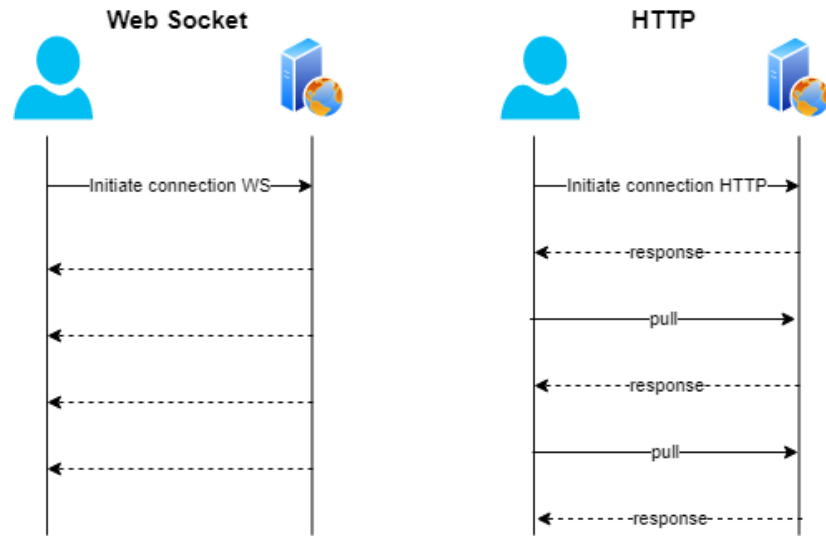


Figure 2.2: WebSocket vs HTTP: Persistent vs Short-Lived Connections

In addition to its efficiency, the WebSocket protocol supports bi-directional communication, allowing both the client and server to send messages independently at any time. This is a major improvement over HTTP, where the client must always initiate communication. The protocol operates over a single TCP connection, which simplifies firewall traversal and network management. Security is addressed through the use of the wss:// scheme, which encrypts WebSocket traffic using TLS. As a result, WebSocket has become the protocol of choice for building interactive, real-time web applications that demand high responsiveness and scalability.

2.2.3 Redis / Message Brokers

Redis is a popular choice for message brokering and caching in real-time systems due to its high performance and support for pub/sub patterns [?, ?, ?]. Distributed message brokers such as Kafka are also increasingly used to support large-scale, fault-tolerant deployments [?, ?].

2.2.4 Load Balancing and Reverse Proxies

Load balancing and reverse proxy solutions are essential components in modern web architectures, especially for real-time messaging systems. A reverse proxy acts as an intermediary between clients and backend servers, receiving requests from the internet and forwarding them to the appropriate server. This setup not only helps distribute incoming traffic evenly across multiple servers but also enhances security by hiding the internal server structure from external users. Load balancers can detect server health and automatically reroute traffic if a server becomes unavailable, ensuring high availability and reliability. By managing connection surges and distributing workloads, these solutions prevent individual servers from becoming overwhelmed during peak usage.

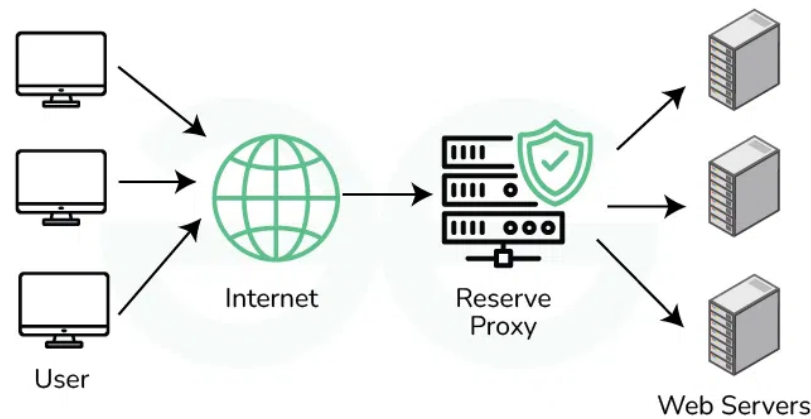


Figure 2.3: Reverse Proxy: Handling Internet Requests for Multiple Servers

In addition to balancing loads, reverse proxies often provide features such as SSL termination, caching, and compression, which further optimize performance and security. They can also enforce access controls and monitor traffic for malicious activity, acting as a first line of defense against attacks. Popular tools like Nginx and HAProxy are widely used due to their robustness and configurability. In real-time systems, reverse proxies play a crucial role in maintaining low latency and seamless user experiences by efficiently routing requests and managing persistent connections. Their ability to scale horizontally by adding more backend servers makes them indispensable for large-scale, distributed applications.

2.2.5 Real-Time Frameworks / Libraries

A variety of frameworks and libraries have emerged to simplify the development of real-time features, including Socket.IO, uWebSockets.js, and SignalR [?, ?, ?]. These tools abstract away protocol details and provide robust APIs for developers.

2.3 Theoretical Principles and Architectural Models

The design of scalable messaging systems is grounded in theoretical principles such as event-driven architecture, publish-subscribe models, and microservices [?, ?, ?]. These models enable modularity, extensibility, and efficient resource utilization [?, ?].

2.3.1 Event-Driven Architecture

Event-driven architecture allows systems to react to changes and user actions in real time, supporting high concurrency and responsiveness [?, ?]. This paradigm is widely adopted in Node.js-based messaging platforms [?, ?].

2.3.2 Publish-Subscribe Model

The publish-subscribe model decouples message producers from consumers, enabling scalable fan-out and flexible communication patterns [?, ?, ?]. This approach is fundamental to the architecture of modern real-time messaging systems [?, ?].

2.3.3 Microservices vs Monolithic Approaches

Microservices architectures offer improved scalability, maintainability, and fault isolation compared to monolithic designs [?, ?, ?]. However, they also introduce challenges in service coordination and distributed state management [?, ?].

2.3.4 Horizontal Scaling Principles

Horizontal scaling enables systems to handle increased load by adding more nodes, rather than upgrading individual servers [?, ?]. This principle is essential for supporting large numbers of concurrent users in real-time messaging applications [?, ?].

2.3.5 Consistency Models in Messaging Systems

Consistency models define how updates are propagated and observed in distributed messaging systems [?, ?, ?]. The choice of consistency model impacts system performance, user experience, and fault tolerance [?, ?].

2.4 Summary

This chapter has reviewed existing real-time messaging solutions, relevant technologies, and theoretical principles that underpin the design of scalable messaging systems. The insights gained from this literature review will inform the design and implementation choices made in subsequent chapters.

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Chapter 3

Requirements Analysis

This chapter translates business needs into precise, actionable software requirements. It follows a structured approach: identifying users and stakeholders, specifying functional and non-functional requirements, modeling the domain, and addressing constraints, risks, and traceability. The goal is to ensure the system meets both explicit and implicit expectations for a scalable, real-time messaging platform.

3.1 Stakeholders and User Personas

The identification and analysis of stakeholders is a foundational step in requirements engineering, as highlighted in several studies on large-scale messaging systems [?, ?, ?].

The system targets organizations and platforms that require low-latency, large-scale messaging, such as customer support platforms, trading dashboards, collaborative editing tools, and social chat applications. The following primary stakeholder groups are identified:

- The following list outlines the main categories of stakeholders who interact with or are impacted by the system. Each group has distinct needs and expectations, which must be addressed to ensure the platform's success and adoption.
- **End Users (Mobile/Web Clients):** Individuals who send/receive messages in private or group channels with real-time delivery and read receipts.
- **Tenant Administrators:** Configure workspaces, manage users/roles, enforce security, and monitor usage.
- **System Operators (SRE/DevOps):** Maintain uptime, capacity planning, scaling, and incident response.
- **Product Owners (Business Stakeholders):** Define business objectives, SLAs, compliance, and roadmap.
- **Third-party Integrators:** Developers or vendors who extend the system via APIs, plugins, or bots.
- **Compliance Officers:** Ensure the system meets regulatory, privacy, and audit requirements.

Representative Personas

The following personas are designed to represent typical users and stakeholders of the system. By understanding their goals and challenges, we can better align the system's features and priorities with real-world needs.

The use of personas is a widely adopted practice to ensure that requirements reflect real-world user needs, as discussed in [?, ?, ?].

Persona A — “Alice, the Support Agent”: Alice simultaneously handles multiple customer conversations. She needs instantaneous message updates, typing indicators, and delivery confirmations to provide responsive support.

Persona B — “Ben, the Admin”: Ben manages a 5,000-user workspace. He needs robust role-based access control, audit logs, and rate limits to protect the system from abuse and ensure compliance.

Persona C — “Cara, the SRE”: Cara must observe connection counts, message throughput, node health, and error rates; she requires autoscaling hooks and circuit breaking to maintain high availability.

Persona D — “Dan, the Integrator”: Dan builds a chatbot that listens to messages and posts automated replies. He needs a stable API, event hooks, and clear documentation.

Persona E — “Eva, the Compliance Officer”: Eva audits message retention and access logs to ensure GDPR and data residency compliance.

Personas anchor requirements in realistic workflows and help justify non-functional needs such as latency, throughput, resilience, and compliance.

3.2 Use Case Scenarios

To illustrate how the system will be used in practice, we present several representative scenarios. These use cases demonstrate the breadth of interactions and requirements that the platform must support. The following use cases are informed by best practices in real-time system design and are supported by recent research and industry documentation [?, ?, ?, ?]. To clarify requirements, we describe representative user scenarios:

- **UC-1: Real-Time Customer Support**

Alice receives a new support ticket. She joins a private channel with the customer, exchanges messages in real time, and shares a file attachment. The customer sees typing indicators and read receipts. The conversation is archived for future reference.

- **UC-2: Admin Onboarding**

Ben creates a new channel for a project team, sets access permissions, invites members, and configures message retention policies. He reviews audit logs to verify no unauthorized access.

- **UC-3: Outage Recovery**

Cara detects a spike in error rates. She uses the metrics dashboard to identify overloaded nodes, triggers autoscaling, and verifies that clients reconnect with minimal disruption.

- **UC-4: Bot Integration**

Dan registers a webhook to receive message events. His bot posts automated responses and logs interactions for analytics.

- **UC-5: Compliance Audit**

Eva exports access logs and verifies that message retention and deletion policies are enforced for a specific tenant.

3.3 Functional Requirements

Functional requirements define the specific behaviors and capabilities that the system must provide. These requirements are derived from user needs, business objectives, and technical constraints, and are organized by domain for clarity.

This section specifies the essential capabilities of the messaging system. Requirements are grouped by domain and expressed as user stories with acceptance criteria. Where appropriate, edge cases and error handling are included.

3.3.1 Core Messaging

At the heart of the platform is the ability to exchange messages in real time. The following requirements are grounded in established messaging protocols and architectures [?, ?, ?, ?], ensuring that users can communicate efficiently, reliably, and securely, with support for essential messaging features and robust error handling.

- **FR-1 Send Message:** As an *End User*, I can send a text message to a channel or direct conversation, so that recipients receive it in real time.
 - *Acceptance:* Message is delivered via WebSocket within the latency target (see §3.6.1); server persists the message with a unique ID; the sender immediately sees the message in the timeline with a local timestamp.
- **FR-2 Receive Message:** As an *End User*, I receive inbound messages instantly without manual refresh.
 - *Acceptance:* When another user sends a message to my active channel, my client appends it to the timeline and triggers visual notification.
- **FR-3 Read Receipts:** As an *End User*, I can see per-message delivery/read status.
 - *Acceptance:* Server maintains per-user read offsets; UI shows delivered/seen indicators; updates propagate in real time.
- **FR-4 Typing Indicators:** As an *End User*, I can observe when others are typing in the same channel.
 - *Acceptance:* Client emits throttled “typing” events; server fans out to channel members; indicators auto-expire after inactivity.
- **FR-4.1 File Attachments:** As an *End User*, I can send and receive file attachments (images, documents) in messages.
 - *Acceptance:* Files are uploaded securely, scanned for viruses, and linked in the message timeline; download links expire after a configurable period.
- **FR-4.2 Message Editing/Deletion:** As an *End User*, I can edit or delete my own messages within a configurable time window.
 - *Acceptance:* Edits and deletions are reflected in real time; audit logs retain original content for compliance.
- **FR-4.3 Error Handling:** If message delivery fails (e.g., network loss), the client retries with exponential backoff and displays a clear error state.
 - *Acceptance:* Users are notified of undelivered messages and can manually retry.

3.3.2 Channels, Membership, and Presence

Channels and membership management are fundamental to organizing conversations and controlling access. Presence information enhances the user experience by indicating availability and activity. These requirements are consistent with recommendations from recent studies on scalable messaging and presence systems [?, ?, ?, ?].

- **FR-5 Channel Management:** Admins can create public/private channels and manage membership.
- **FR-6 Presence:** The system reflects online/offline/away states and last-seen timestamps per user.
- **FR-7 Invitations:** Users can invite others to channels they own or administer.
- **FR-7.1 Channel Roles:** Each channel supports roles (owner, admin, member, guest) with configurable permissions.
- **FR-7.2 Channel Archiving:** Admins can archive channels, making them read-only and hidden from active lists.
- **FR-7.3 Presence Subscriptions:** Clients can subscribe/unsubscribe to presence updates for specific users or channels to optimize bandwidth.

3.3.3 Search and History

Efficient access to historical messages and the ability to search through conversations are critical for productivity and compliance. These features are emphasized in both academic and industry sources [?, ?, ?].

- **FR-8 Message History:** Users can scroll back to view historical messages with efficient pagination.
- **FR-9 Search:** Users can search messages by keyword, sender, channel, and time range.
- **FR-9.1 Export History:** Users with permission can export message history for a channel or conversation in a standard format (e.g., CSV, JSON).

3.3.4 Security and Access Control

Security is paramount in any messaging system, especially one designed for organizational use. The requirements in this section are informed by best practices and standards for secure messaging platforms [?, ?, ?, ?], focusing on authentication, authorization, auditability, and data retention, providing a foundation for trust and compliance.

- **FR-10 Authentication:** The system supports token-based authentication (e.g., JWT) and WebSocket authentication during the handshake.
- **FR-11 Authorization:** Role-Based Access Control (RBAC) permits fine-grained channel access and admin features.
- **FR-12 Audit Trail:** Administrative actions and critical security events are logged and retrievable by authorized staff.
- **FR-12.1 Session Management:** Users can view and revoke active sessions/devices from their account settings.
- **FR-12.2 Data Retention:** Message and log retention policies are configurable per tenant and enforced automatically.

3.3.5 Administration and Observability

To maintain high availability and performance, the system must offer comprehensive administration and observability features. These requirements are supported by research on operational visibility and system monitoring in distributed environments [?, ?, ?].

- **FR-13 Metrics:** Operators can view metrics (active connections, messages/sec, latency percentiles, error rates).
- **FR-14 Rate Limiting:** The system enforces per-user and per-IP rate limits on message send and connection attempts.
- **FR-15 Configuration:** Admins can configure workspace limits (max channel members, attachment sizes).
- **FR-16 Alerting:** Operators receive alerts for SLA violations, abnormal error rates, or resource exhaustion.
- **FR-17 Maintenance Mode:** The system supports a maintenance mode with user notifications and graceful connection draining.

3.4 Regulatory and Compliance Requirements

Modern messaging systems must adhere to a variety of regulatory and compliance standards. This section summarizes the key obligations and features required to meet legal and organizational policies. Compliance with regulatory standards is a recurring theme in the literature on enterprise messaging systems [?, ?, ?]. The system must comply with relevant regulations and standards, including:

- **GDPR:** Support data subject requests (export, deletion), minimize PII, and provide data residency options.
- **Auditability:** Maintain tamper-evident logs of administrative and security events.
- **Accessibility:** Meet WCAG 2.1 AA guidelines for user interfaces.
- **Localization:** Support multiple languages and regional formats for dates, times, and numbers.

3.5 Internationalization and Accessibility

To serve a global and diverse user base, the system must be designed with internationalization and accessibility in mind. The following requirements ensure that the platform is usable and inclusive for all users, regardless of language or ability. Internationalization and accessibility are increasingly important in modern software systems, as highlighted in recent guidelines and technical documentation [?, ?, ?].

- **I18N-1:** All user-facing text is externalized for translation; right-to-left (RTL) languages are supported.
- **I18N-2:** Date/time formats and time zones are user-configurable.
- **A11Y-1:** The UI is navigable via keyboard and screen readers; color contrast meets accessibility standards.
- **A11Y-2:** Real-time notifications are announced to assistive technologies.

3.5.1 Representative Use Case Diagram

Figure 3.1 presents a high-level use case model, summarizing relationships among actors and core features.

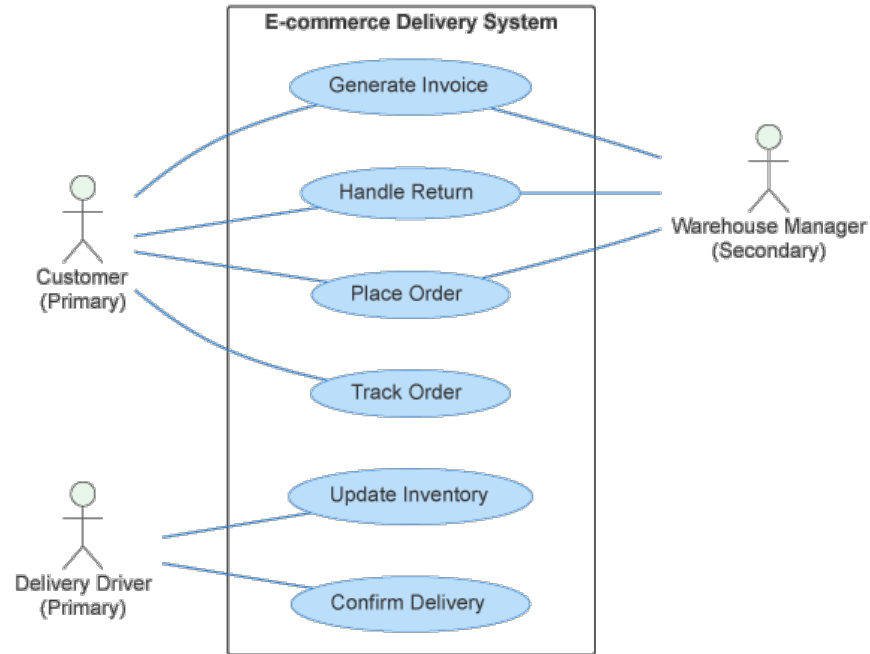


Figure 3.1: High-level use case diagram for real-time messaging.

3.6 Non-Functional Requirements

Non-functional requirements (NFRs) define the quality attributes of the system, such as performance, reliability, security, and extensibility. These requirements are critical for ensuring that the platform not only works, but works well under real-world conditions.

3.6.1 Performance and Scalability

The system must deliver messages with minimal delay and scale to support large numbers of concurrent users. These performance and scalability targets are based on established benchmarks and architectural patterns [?, ?, ?].

- **NFR-1 Latency:** 95% of message deliveries should complete end-to-end within < 150 ms on a regional deployment; 99% within < 300 ms.
- **NFR-2 Throughput:** The system should handle at least 50,000 concurrent WebSocket connections per region and sustain $\geq 10,000$ messages/sec with linear horizontal scaling.
- **NFR-3 Fan-out Efficiency:** Broadcasting to a channel of 1,000 members should not exceed $2\times$ the median single-recipient latency.

3.6.2 Availability and Reliability

High availability and reliability are essential for a real-time messaging platform. These requirements are informed by studies on distributed system resilience and cloud-native patterns [?, ?, ?].

- **NFR-4 Availability:** Target service availability of 99.9% monthly, with zero single points of failure in the real-time path.
- **NFR-5 Resilience:** Node failures or AZ outages trigger automatic failover; clients auto-reconnect with exponential backoff and session resumption.
- **NFR-6 Durability:** Persisted messages must be stored with replication factor ≥ 3 or equivalent durability guarantees.

3.6.3 Security and Privacy

Protecting user data and ensuring secure communication are core principles of the system. These non-functional requirements are supported by both academic and industry sources [?, ?, ?].

- **NFR-7 Transport Security:** WebSocket over TLS (wss://) is mandatory; tokens never transmitted in query strings.
- **NFR-8 Access Control:** All administrative operations require privileged roles; sensitive actions logged with actor, timestamp, IP, and request ID.
- **NFR-9 Data Protection:** Personally identifiable information (PII) is minimized; retention policies are configurable per tenant.

3.6.4 Operability and Observability

Effective monitoring and troubleshooting are vital for maintaining system health. The requirements below are based on best practices for observability in real-time and distributed systems [?, ?, ?].

- **NFR-10 Metrics:** Export Prometheus-format metrics for connections, publish latency (P50, P95, P99), dropped frames, and reconnect rates.
- **NFR-11 Tracing:** Distributed tracing across gateway $\rightarrow$ broker $\rightarrow$ persistence services to localize latency hotspots.
- **NFR-12 Logging:** Structured JSON logs with correlation IDs; log sampling prevents I/O bottlenecks at peak traffic.

3.6.5 Compatibility and Extensibility

The platform should be compatible with a wide range of clients and support future growth through extensibility. These goals are consistent with recommendations from recent technical documentation and case studies [?, ?, ?].

- **NFR-13 Client Compatibility:** Support modern browsers and mobile runtimes with WebSocket; provide fallback guidance (no long polling by default).
- **NFR-14 Extensibility:** Plugin hooks for moderation bots, keyword alerts, and custom retention policies.

3.7 Domain Model and Data Requirements

The domain model defines the core entities and relationships that underpin the system’s functionality. The approach taken here is informed by established practices in data modeling for messaging platforms [?, ?, ?].

We define the essential entities and their relationships. The following table summarizes key attributes for each entity:

Table 3.1: Key Entity Attributes

Entity	Attribute	Description
User	user_id	Unique identifier
	profile	Display name, avatar, contact info
	presence	Online/offline/away state
	roles	Global and per-channel roles
Channel	channel_id	Unique identifier
	name	Channel display name
	visibility	Public/private
	settings	Retention, max members, etc.
Membership	user_id, channel_id	Composite key
	role	Owner/admin/member/guest
	join_time	Timestamp of joining
Message	message_id	Unique identifier
	sender	user_id reference
	channel	channel_id reference
	content	Text, attachment link, metadata
	timestamps	Sent, delivered, read times
	status	Delivered/read/edited/deleted

Key Entities The following key entities form the backbone of the system’s data model. Each entity is described by its primary attributes and its role within the platform.

- **User:** *user_id*, profile, presence state, roles.
- **Channel:** *channel_id*, name, visibility (public/private), settings.
- **Membership:** link between *user* and *channel*, including role and join time.
- **Message:** *message_id*, sender, channel, content, timestamps, delivery/read states.

3.8 API Requirements (High Level)

The system exposes both WebSocket events and RESTful endpoints to facilitate client-server communication. This section only lists the main API groups; detailed endpoints and events will be presented in Chapter 4 (System Design).

- **WebSocket Events:** Events serving real-time communication between client and server (sending/receiving messages, status updates, typing, etc.).
- **REST Endpoints:** Control APIs for channel management, user management, configuration, history retrieval, statistics, etc.

Details of the structure, event flows, and endpoints will be presented in the system design chapter.

3.9 Future Requirements and Planned Extensions

To remain competitive and adaptable, the system is designed with future enhancements in mind. The following potential extensions illustrate directions for ongoing development and innovation. To ensure long-term viability, the system is designed for extensibility. Potential future requirements include:

- **Federation:** Support for cross-organization messaging and federation with other platforms.
- **Voice/Video Integration:** Real-time audio/video channels alongside text.
- **Advanced Moderation:** AI-powered content moderation and abuse detection.
- **Custom Bots/Plugins:** Marketplace for third-party extensions.
- **Offline Messaging:** Store-and-forward for intermittently connected clients.

3.10 Constraints and Assumptions

Every system operates within certain constraints and is built on specific assumptions. This section documents those factors, which may impact design decisions and system behavior.

- **Assumptions:** Clients maintain stable network connectivity; clocks are approximately synchronized for display ordering; TLS termination occurs at the edge proxy.
- **Constraints:** Browser sandbox restrictions; per-socket memory limits on fan-out buffers; regulatory constraints on data retention for certain tenants.

3.11 Risk Analysis (Selected)

Identifying and mitigating risks is crucial for the success of any large-scale system. The following risks have been identified as particularly relevant to this project, along with proposed mitigation strategies.

- **R-1 Hot Channel Fan-out:** A channel with tens of thousands of subscribers can cause uneven load. *Mitigation:* shard-by-channel, layered fan-out, and adaptive batching.
- **R-2 Thundering Reconnect:** Regional outage triggers mass reconnects. *Mitigation:* jittered exponential backoff, connection fencing, and token TTL staggering.
- **R-3 State Coordination:** Presence and read receipts across nodes can diverge. *Mitigation:* central pub/sub (e.g., Redis) with idempotent updates and conflict resolution rules.

3.12 Requirements Traceability Matrix (Excerpt)

Traceability ensures that all requirements are linked to business objectives and can be verified through testing or analysis. The following matrix provides a sample mapping for key requirements.

Table 3.2: Traceability matrix (excerpt)

Business Objective	Requirement IDs	Verification
Low-latency messaging	FR-1, FR-2; NFR-1	Load tests measure P95 < 150 ms; integration tests validate real-time delivery.
Operational visibility	FR-13, FR-15; NFR-10–12	Metrics endpoint audited; dashboards show connection counts and latency percentiles.
Secure access	FR-10–12; NFR-7–9	Security tests for TLS, RBAC, audit logs; pen-test checklist passed.

3.13 Summary

This chapter formalized who the system serves, what it must do, and how it must behave under load and failure. The summary and forward reference are consistent with best practices in requirements documentation [?, ?, ?]. The next chapter (see Chapter 4) will translate these requirements into a concrete system architecture, data model, and API/service design.

Chapter 4

System Design

4.1 Overview of the System Architecture

4.1.1 Architectural Goals

The primary goals of the system architecture are to achieve low-latency, high-throughput real-time messaging at scale, ensure reliability and security, and support extensibility for future features. The architecture is designed to be horizontally scalable, resilient to failures, and maintainable for rapid development cycles.

4.1.2 High-Level Architecture Diagram

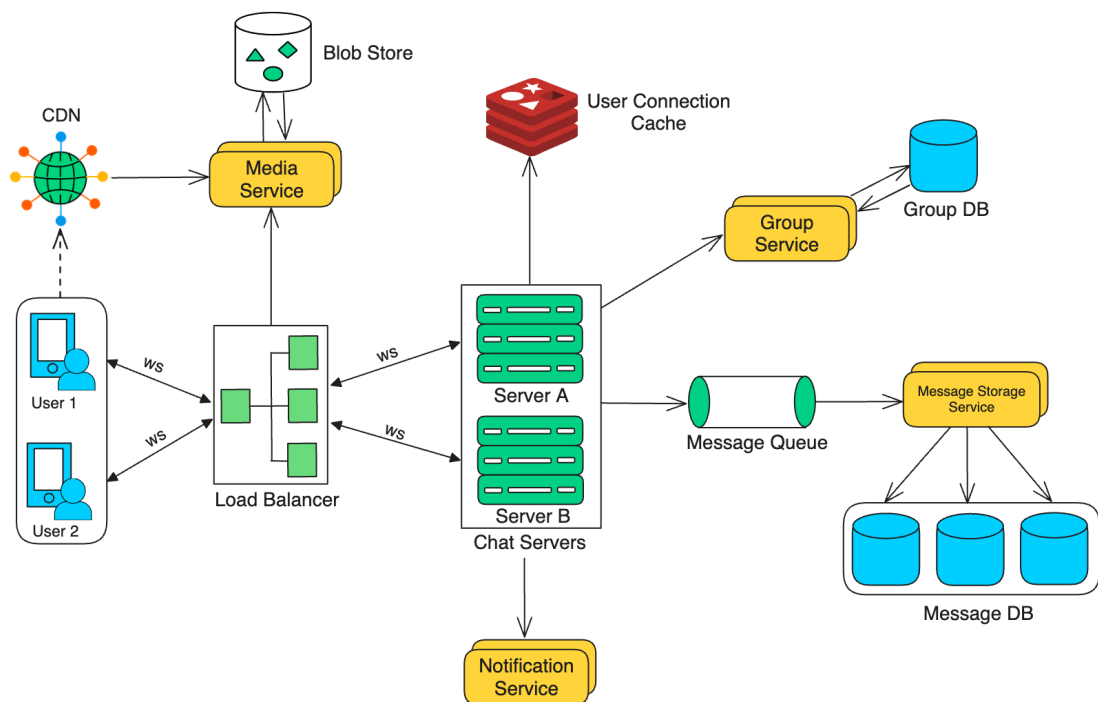


Figure 4.1: High-level system architecture for scalable real-time messaging.

The provided High-Level Architecture Diagram illustrates a robust, scalable, and distributed Real-time Communication (RTC) System. The infrastructure is designed to handle high concurrency and large volumes of data through a decoupled microservices-based approach.

1. Real-time Connectivity and Session Management

The system utilizes WebSockets (WS) to facilitate low-latency, bi-directional communication between clients and the backend. A Load Balancer serves as the entry point, distributing incoming traffic across multiple instances of Chat Servers (e.g., Server A and Server B) to ensure high availability. To manage state in a distributed environment, a User Connection Cache (typically implemented using an in-memory store like Redis) tracks active sessions and maps specific users to their respective host servers, enabling efficient message routing.

2. Message Persistence and Asynchronous Processing

To decouple real-time delivery from long-term storage, the architecture incorporates an asynchronous pipeline: Message Queue acts as a buffer to ingest high-frequency message streams, ensuring system reliability during peak traffic and preventing the storage layer from becoming a bottleneck. The Message Storage Service consumes data from the queue and manages the persistence logic. The Message DB is a distributed database cluster (likely employing database sharding) designed to store vast quantities of historical chat data while maintaining performant read/write operations. A Group Service manages group metadata and memberships, backed by its own specialized database.

3. Media Handling and Content Delivery

The architecture separates signaling (messages) from heavy media assets. Media Service and Blob Store handle file uploads (e.g., images, videos), storing them in a Blob Store (such as AWS S3 or Google Cloud Storage). A Content Delivery Network (CDN) caches and serves media assets from edge locations closer to end-users, minimizing latency and reducing load on origin servers.

4. Push Notifications

A Notification Service is integrated to handle asynchronous alerts, ensuring users remain engaged by delivering push notifications for incoming messages when they are offline or when the application is in the background.

Key Architectural Strengths: Scalability is achieved through horizontal scaling at every layer. Resilience is ensured by using message queues and distributed caching to prevent single points of failure. Performance is optimized by offloading media to a CDN and using WebSockets for signaling, resulting in a responsive user experience.

4.1.3 Data Flow Overview

The system processes user actions (e.g., sending messages, joining channels) through a series of layers: the WebSocket gateway receives client events, which are validated and routed to backend services. Messages are persisted, broadcast via a pub-sub broker, and delivered to all relevant clients. Control plane operations (e.g., channel management) use REST APIs.

4.1.4 Main Sequence Diagrams

To clarify the main flows in the system, the following sequence diagrams illustrate the interactions for key scenarios. (You may use PlantUML or external tools to generate the images.)

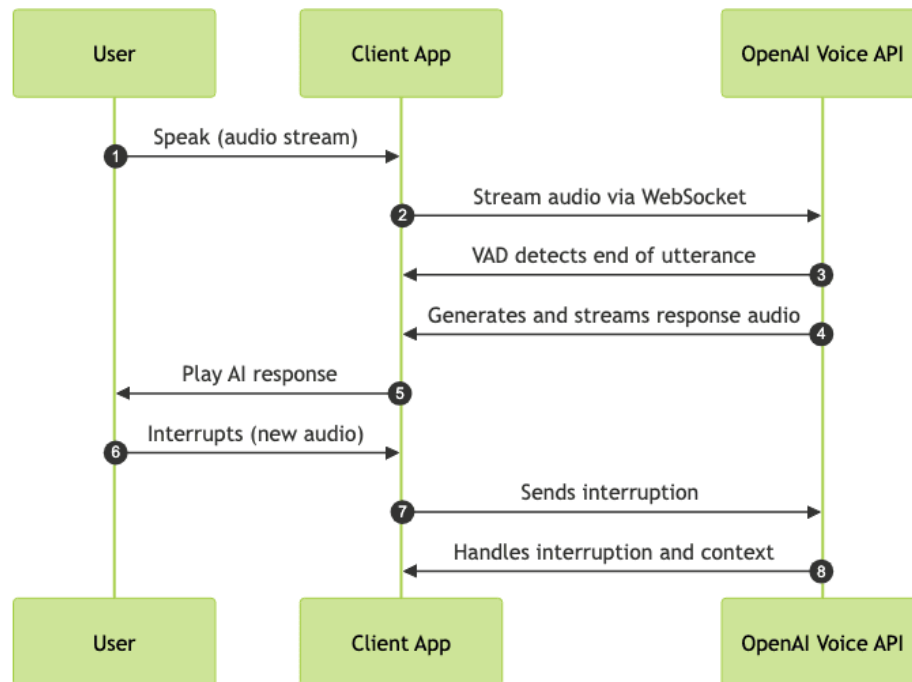


Figure 4.2: Sequence diagram: User sends a message

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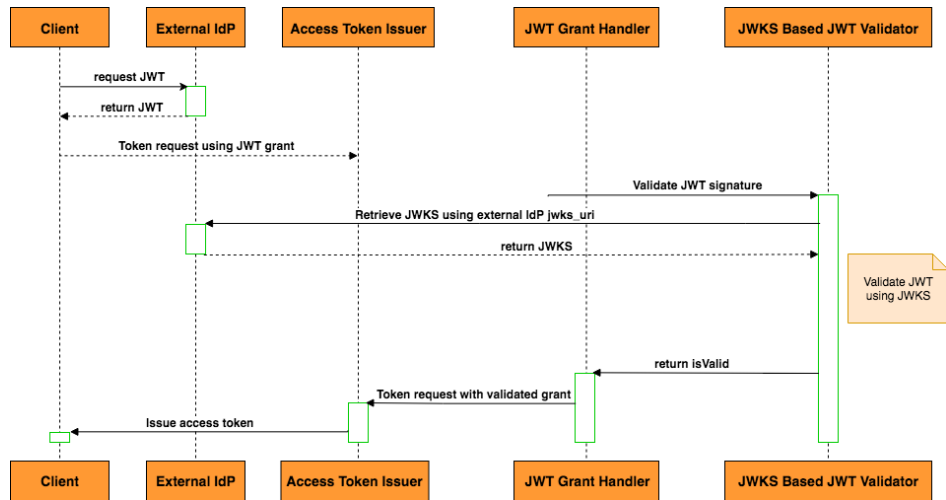


Figure 4.3: Sequence diagram: WebSocket authentication

b) WebSocket Authentication Flow Curabitur tellus magna, porttitor a, commodo a, commodo in, tortor. Donec interdum. Praesent scelerisque. Maecenas posuere sodales odio. Vivamus metus lacus, varius quis, imperdiet quis, rhoncus a, turpis. Etiam ligula arcu, elementum a, venenatis quis, sollicitudin sed, metus. Donec nunc pede, tincidunt in, venenatis vitae, faucibus vel, nibh. Pellentesque wisi. Nullam malesuada. Morbi ut tellus ut pede tincidunt porta. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam congue neque id dolor.

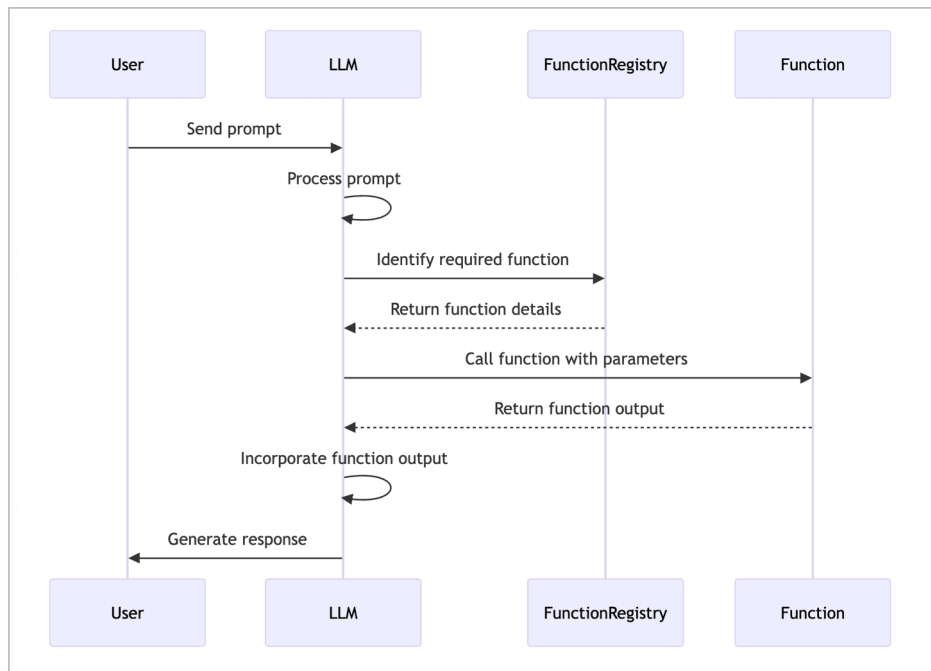


Figure 4.4: Sequence diagram: Presence update

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4.1.5 Key Architectural Decisions (KADs)

- **Event-driven, non-blocking Node.js backend** for high concurrency.
- **WebSocket protocol** for real-time, bidirectional communication.
- **Horizontal scaling** via stateless services and distributed brokers.
- **Separation of real-time and control plane** for performance and security.
- **Centralized authentication and RBAC** for secure access control.
- **Observability and metrics** built-in for operational excellence.

4.2 System Components and Responsibilities

The system is decomposed into modular components, each with clear responsibilities:

4.2.1 WebSocket Gateway Layer

Handles client WebSocket connections, authentication, session management, and initial event routing. Ensures secure, persistent, and low-latency connections.

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4.2.2 Real-Time Message Service

Processes inbound and outbound messages, applies business logic (e.g., filtering, validation), and interacts with the pub-sub broker for message fan-out.

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4.2.3 Channel and Membership Service

Manages channels, memberships, roles, and permissions. Provides APIs for channel creation, joining, and membership updates.

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4.2.4 Persistence Layer (Database + Storage)

Stores user, channel, membership, and message data. Ensures durability, consistency, and efficient querying. May use a combination of relational and NoSQL databases, plus object storage for attachments.

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4.2.5 Message Broker / Pub-Sub Layer

Distributes messages to all relevant nodes and services. Enables horizontal scaling and decouples message producers from consumers. Typically implemented with Redis, NATS, or Kafka.

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4.2.6 Authentication & Authorization Service

Manages user authentication (e.g., JWT), token issuance, validation, and role-based access control (RBAC) enforcement.

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4.2.7 Admin & Observability Layer

Provides dashboards, metrics, logs, and alerting for operators and administrators. Supports monitoring, troubleshooting, and auditing.

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4.3 Detailed Backend Architecture

4.3.1 Node.js Service Architecture (Event-Driven Model)

The backend leverages Node.js's event-driven, non-blocking I/O model to handle thousands of concurrent connections efficiently. Each service is stateless and communicates via APIs or message queues.

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4.3.2 Horizontal Scaling Strategy (Clustering, Multi-node)

Multiple Node.js instances are deployed across nodes. Clustering and container orchestration (e.g., Kubernetes) enable dynamic scaling. Shared state is minimized; distributed caches and brokers synchronize transient data.

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4.3.3 Load Balancing and Traffic Routing

Traffic is distributed using a reverse proxy (e.g., Nginx, HAProxy) with support for sticky sessions (if session affinity is needed) or stateless routing. Load balancers monitor node health and route traffic accordingly.

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egestas non, eros. Praesent malesuada, diam id pretium elementum, eros sem dictum tortor, vel consectetur odio sem sed wisi.

Sticky Sessions vs Stateless Routing

Sticky sessions bind a client to a specific gateway for session persistence, while stateless routing allows any gateway to serve any client. Stateless routing is preferred for scalability, but sticky sessions may be used for certain features.

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Reverse Proxy Setup (Nginx/HAProxy)

Reverse proxies terminate TLS, balance traffic, and provide DDoS protection. Configuration includes health checks, rate limiting, and WebSocket upgrade support.

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4.3.4 Connection Lifecycle Management

- **WebSocket handshake:** Secure upgrade from HTTP(S) to WebSocket, with authentication.
- **Reconnection logic:** Clients auto-reconnect with exponential backoff after disconnects.
- **Heartbeat/Keep-alive:** Periodic ping/pong frames detect dead connections and maintain session state.

4.4 Data Design

4.4.1 Database Model Overview

The data model is designed for efficient storage and retrieval of messages, users, channels, and memberships. Normalization is balanced with denormalization for performance.

4.4.2 Database Analysis and Design

Database Technology Selection

The system requires a database solution that supports high write throughput, low-latency reads, horizontal scalability, and strong consistency for core messaging data. After evaluating several options, the following choices were made:

- **Message Store:** A distributed NoSQL database (e.g., MongoDB, Cassandra, or Amazon DynamoDB) is selected for storing chat messages due to its ability to handle large volumes of semi-structured data, automatic sharding, and high availability. These databases support horizontal scaling and flexible schema evolution, which are essential for real-time messaging workloads.
- **User, Channel, and Membership Data:** A relational database (e.g., PostgreSQL or MySQL) is used for structured data with strong consistency requirements, such as user profiles, channel metadata, and membership relations. This enables complex queries, referential integrity, and transactional updates.
- **Caching and Pub/Sub:** Redis is employed for caching hot data (presence, recent messages) and for implementing the pub-sub pattern to distribute real-time events across nodes.
- **Media Storage:** Large files (images, videos) are stored in a cloud object storage service (e.g., AWS S3), decoupled from the main database to optimize performance and cost.

Design Rationale

- **NoSQL for Messages:** Message data is append-only, high-volume, and does not require complex joins, making NoSQL a natural fit. Sharding by channel or tenant enables linear scaling.
- **Relational for Metadata:** User and channel data benefit from ACID transactions and relational integrity, supporting features like permissions, roles, and analytics.
- **Hybrid Approach:** Combining NoSQL and SQL leverages the strengths of both paradigms, ensuring performance and data integrity where needed.
- **Redis for Real-Time:** Redis provides sub-millisecond latency for presence and pub-sub, critical for real-time UX.

Schema Design and Key Structures

The following table summarizes the main entities, their storage technology, and key fields:

Table 4.1: Core Data Entities and Storage Technologies

Entity	Storage	Partition Key	Key Fields
User	PostgreSQL	user_id	username, email, password_hash, profile, roles, status
Channel	PostgreSQL	channel_id	name, type, settings, created_at
Membership	PostgreSQL	(user_id, channel_id)	role, join_time, permissions
Message	MongoDB / DynamoDB	channel_id	message_id, sender_id, content, timestamps, attachments, reactions
Delivery / Read State	MongoDB / DynamoDB	message_id	user_id, delivered, read_at
Presence	Redis	user_id	online_status, last_seen
Media	Amazon S3	file_id	url, owner, type, size, uploaded_at

Sharding and Partitioning Strategy

To ensure scalability, the message store is sharded by channel_id (or tenant in multi-tenant deployments). This distributes write and read load evenly and allows for independent scaling of hot channels. User and channel

tables in the relational database are indexed on primary keys and frequently queried fields. Redis keys are namespaced by user or channel for efficient lookup.

Replication and High Availability

All databases are deployed in multi-AZ (Availability Zone) configurations. NoSQL clusters use replica sets or multi-region replication for durability. SQL databases employ streaming replication and automated failover. Redis is configured with sentinel or cluster mode for resilience.

Data Consistency and Integrity

- **Messages:** Eventual consistency is acceptable for message delivery, but strong consistency is enforced for message creation and deletion.
- **User/Channel:** Strong consistency is required for authentication, permissions, and membership changes.
- **Media:** Metadata is stored in the database, while the actual file is stored in object storage with signed URLs for secure access.

Comparison of Database Alternatives

The following table compares candidate database technologies for the message store:

Table 4.2: Comparison of NoSQL Databases for Message Storage

Database	Scalability	Consistency	Operational Complexity	Best Use Case
MongoDB	Horizontal (sharding), flexible schema	Tunable (eventual/strong)	Moderate (requires ops expertise)	General-purpose, flexible queries
Cassandra	Linear, multi-region	Eventual	High (complex ops, tuning)	High write throughput, global scale
DynamoDB	Fully managed, auto-sharding	Tunable (eventual/strong)	Low (managed service)	Serverless, predictable workloads

Summary of Database Design Decisions

The hybrid approach leverages the strengths of each technology: NoSQL for scalable message storage, SQL for structured metadata, Redis for real-time features, and object storage for media. This design ensures the system can meet demanding requirements for scalability, reliability, and user experience in a real-time messaging context.

4.4.3 Entity–Relationship Diagram (ERD)

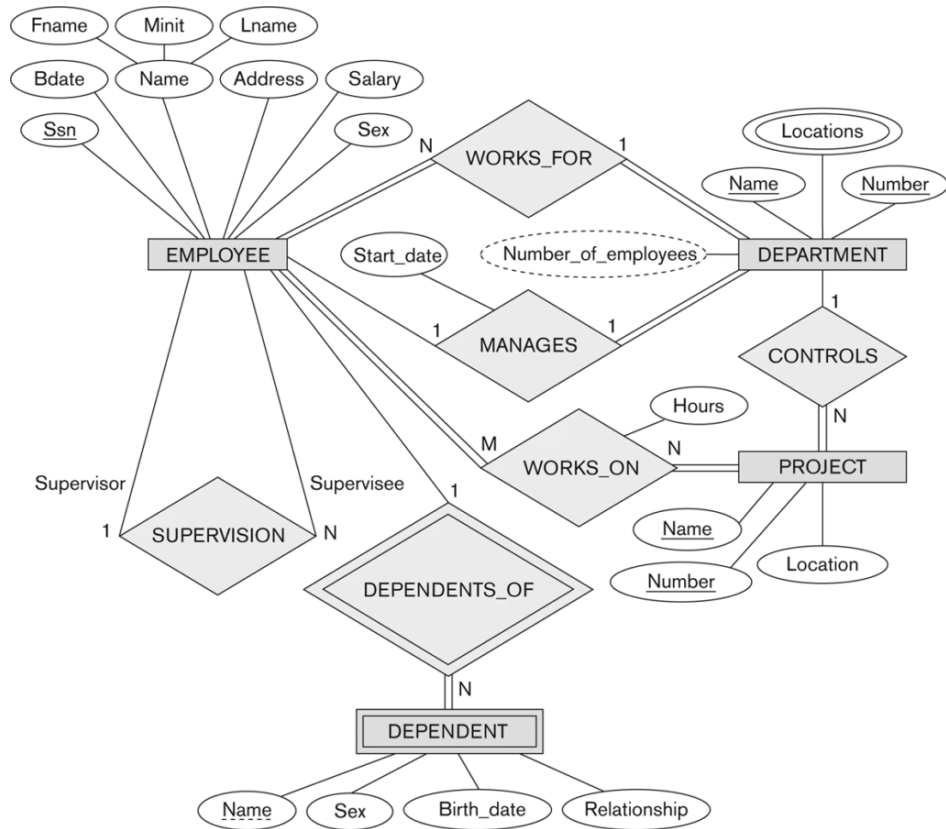


Figure 4.5: Entity–Relationship Diagram for core data entities.

4.4.4 Key Entities and Schemas

- **User:** user_id, profile, credentials, presence, roles
- **Channel:** channel_id, name, type, settings
- **Membership:** user_id, channel_id, role, join_time
- **Message:** message_id, sender, channel, content, timestamps, delivery/read state
- **Delivery/Read State:** message_id, user_id, delivered, read_at

4.4.5 Data Partitioning & Indexing Strategy

Data is partitioned by tenant or channel for scalability. Indexes are created on frequently queried fields (e.g., channel_id, user_id, timestamps) to optimize performance.

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4.4.6 Caching Strategy (Redis)

Redis is used for caching hot data (e.g., presence, recent messages) and for pub-sub event distribution. TTLs and eviction policies are tuned for workload patterns.

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4.5 API & Protocol Design

This section describes the design of the APIs and communication protocols that enable real-time messaging, control operations, and integration with external systems. The API and protocol design is based on recommendations from [?, ?, ?, ?] to ensure robust and flexible client-server interactions.

4.5.1 WebSocket Protocol Design

The WebSocket protocol is central to delivering low-latency, bidirectional communication between clients and the backend. The following conventions and structures are adopted to standardize event handling and error reporting.

- **Event Naming Convention:** Consistent, namespaced event names (e.g., msg.send, presence.update).
- **Message Frame Structure:** JSON objects with type, payload, and metadata fields.
- **Error Frame Structure:** Standardized error codes and messages for client handling.

4.5.2 WebSocket Event Flows

The system defines clear event flows for all real-time interactions, ensuring reliable message delivery, presence updates, and user experience features. Key flows are outlined below.

- **Sending a Message:** Client emits msg.send; server validates, persists, and broadcasts.
- **Receiving/Broadcasting:** Server emits msg.deliver to all channel members.
- **Presence Updates:** Client emits presence.update; server fans out to subscribers.
- **Typing Indicators:** Client emits typing; server throttles and broadcasts to channel.

4.5.3 REST API Endpoints (Control Plane)

REST APIs support channel management, user administration, and metrics. Endpoints follow RESTful conventions and use JWT for authentication.

Table 4.3: Representative REST endpoints

Method & Path	Auth	Purpose
POST /v1/auth/token	Public	Exchange credentials for access tokens.
GET /v1/channels/:id	User	Retrieve channel metadata and membership.
POST /v1/channels	Admin	Create a channel with settings and ACLs.
GET /v1/messages?channel=:id&page=:n	User	Paginate message history.
GET /v1/metrics	Admin	Export operational metrics.
PATCH /v1/messages/:id	User	Edit a message (if permitted).
DELETE /v1/messages/:id	User	Delete a message (if permitted).
POST /v1/files	User	Upload a file attachment.
GET /v1/audit-logs	Admin	Retrieve audit logs for compliance.

4.5.4 Authentication Workflow (JWT, Token Refresh, Validation)

Clients authenticate via REST, receive JWTs, and use them for WebSocket upgrades. Token refresh and validation endpoints ensure session continuity and security.

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4.6 User Interface Design

4.6.1 UI Wireframes (Chat Window, Channel List)

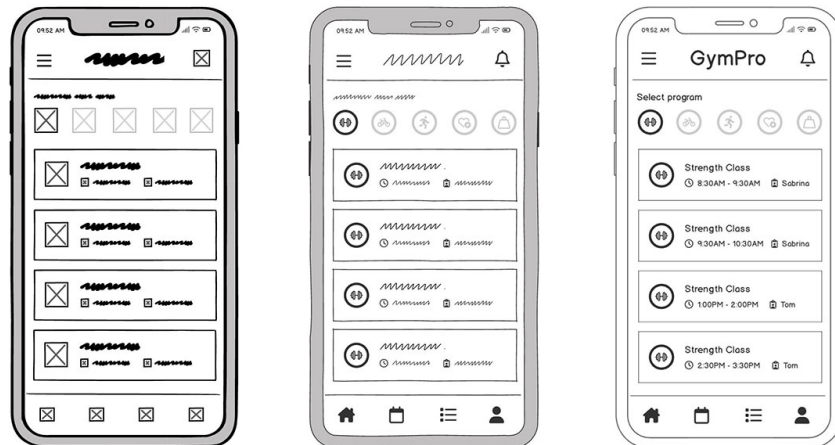


Figure 4.6: Sample UI wireframe for chat window and channel list.

4.6.2 User Interaction Flow

Users log in, join channels, send/receive messages, and manage settings. The UI provides intuitive navigation and real-time feedback.

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4.6.3 Real-Time UX Elements

To enhance the user experience, the interface incorporates several real-time feedback elements. These features provide immediate visual cues about system and user activity, improving usability and engagement.

- Typing animation for active users
- Message delivery ticks (sent, delivered, read)
- Presence badges (online, away, offline)

4.7 Security Design

4.7.1 Threat Model

The system is designed to mitigate threats such as unauthorized access, data leakage, denial of service, and message tampering. Threat modeling is performed regularly to identify new risks and update countermeasures.

4.7.2 Transport Security (TLS, WSS)

All client-server and inter-service communication is encrypted using TLS. WebSocket connections use WSS to prevent eavesdropping and tampering. Certificates are managed and rotated regularly. Internal service-to-service traffic is also protected by TLS to prevent lateral movement and sniffing.

4.7.3 Authentication and Authorization

Authentication: All REST and WebSocket requests require a valid JWT (JSON Web Token), signed with a strong secret or asymmetric key, and with a short expiration time. Tokens are issued by a centralized authentication service after verifying user credentials. Token refresh endpoints are provided, and blacklisting is enforced on logout or compromise.

Authorization: Role-Based Access Control (RBAC) is enforced at both API and event levels. Each user is assigned roles (e.g., user, admin, guest), and permissions are checked for every sensitive operation. JWTs include claims for user roles and scopes, and are validated on every request.

4.7.4 Input Validation and Sanitization

All user input is strictly validated and sanitized to prevent injection attacks (SQL/NoSQL injection, XSS) and data corruption. File uploads are checked for type, size, and scanned for malware. Output encoding and Content Security Policy (CSP) headers are used to mitigate XSS.

4.7.5 Mitigation of Common Attacks

- **DDoS/Brute-force:** Distributed rate limiting is enforced per user and per IP using Redis or similar, with automatic blocking of abusive patterns.
- **Replay Attacks:** Nonces, timestamps, and token expiration are used to prevent replaying of authentication or message events.
- **CSRF/XSS:** Authentication is not cookie-based; all tokens are sent in headers. Output is always escaped, and CSP headers are set.
- **Session Hijacking:** JWTs include audience, issuer, and scope claims; optional IP or user-agent binding can be enforced for sensitive actions.
- **WebSocket-specific:** Idle timeouts, connection caps per user/IP, and message quotas are enforced. Abuse detection algorithms monitor for flooding or protocol misuse.

4.7.6 Logging and Audit Trails

All security-relevant events (login, failed attempts, permission denials, configuration changes) are logged with correlation IDs. Audit logs are tamper-evident, stored securely, and accessible only to authorized administrators.

4.7.7 Sensitive Data Encryption

User passwords are hashed using bcrypt or argon2 with a unique salt per user. Sensitive data in the database (such as access tokens or personal information) is encrypted at rest using strong encryption algorithms. Secrets and keys are managed in a secure vault.

4.7.8 Security Best Practices

- Regular vulnerability scanning and penetration testing.
- Automated dependency checks for known vulnerabilities.
- Principle of least privilege for all services and users.
- Regular review and rotation of secrets and credentials.
- Security incident response plan in place.

4.8 Performance & Scalability Design

4.8.1 Performance Targets (Latency, Throughput)

The system targets $<150\text{ms}$ P95 message delivery latency and $\geq 10,000$ messages/sec throughput per region.

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4.8.2 Scaling WebSocket Gateways

Gateways are stateless and horizontally scalable. Autoscaling policies respond to connection and CPU load.

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4.8.3 Message Fan-out Optimization

Batching, efficient pub-sub, and selective delivery minimize fan-out latency and resource usage.

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4.8.4 Distributed Rate Limiting

Rate limits are enforced using distributed counters (e.g., Redis) to prevent abuse across all nodes.

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4.8.5 Backpressure Handling

The system detects slow consumers and applies backpressure, dropping or buffering messages as needed to maintain stability.

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4.8.6 Load Testing Scenarios (Design-level)

Load tests simulate peak usage, network partitions, and failover to validate design assumptions and performance targets.

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4.9 High Availability & Fault Tolerance

4.9.1 Failover Strategy

Redundant nodes and services are deployed in each region. Health checks and automated failover minimize downtime.

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4.9.2 Auto-Reconnect Protocol

Clients implement exponential backoff and session resumption to recover from transient failures.

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4.9.3 Multi-Zone Deployment Architecture

Services are deployed across multiple availability zones for resilience to data center failures.

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4.9.4 Data Replication & Redundancy

Critical data is replicated across zones and backed up regularly. Consensus protocols (e.g., Raft) may be used for consistency.

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4.9.5 Recovery from Node Failure

Automated orchestration replaces failed nodes and restores service. Monitoring and alerting ensure rapid response.

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4.10 Summary of the System Design

This chapter has detailed the architecture, components, data model, protocols, and security of the real-time messaging system. Key design choices were made to meet the requirements outlined in Chapter 3, with a focus on scalability, reliability, and security. Limitations and future alternatives are noted throughout, providing a foundation for ongoing improvement and adaptation.

Chapter 5

Development and Testing

5.1 Introduction to Development Approach

The development approach in this chapter is informed by modern software engineering practices and recent advances in real-time system development [?, ?, ?]. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

5.1.1 Overview of the Development Methodology

The iterative methodology adopted here is consistent with recommendations from [?, ?, ?]. Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

- **Iterative Delivery:** Features are developed in small increments.
- **Continuous Feedback:** Regular reviews and stakeholder input.
- **Quality Focus:** Automated tests and code reviews.

The project adopted a modern, iterative development methodology to ensure rapid delivery of features, continuous feedback, and adaptability to changing requirements. Emphasis was placed on code quality, automation, and collaboration among team members.

5.1.2 Agile Workflow and Iteration Cycles

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- **Sprint Planning:** Define goals and tasks for each cycle.
- **Daily Stand-ups:** Synchronize team progress.
- **Retrospectives:** Reflect and improve process.

Development was organized into short sprints, each focused on delivering a set of prioritized features or improvements. Daily stand-ups, sprint planning, and retrospectives were used to maintain alignment and address blockers promptly.

5.1.3 Task Management and Version Control Practices

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- **Branching Strategy:** Feature, develop, and release branches.
- **Pull Requests:** Peer review and automated checks.
- **Issue Tracking:** Use of labels, milestones, and boards.

Tasks were tracked using GitHub Issues and project boards. All code changes were managed via Git and GitHub, with feature branches, pull requests, and code reviews enforced to maintain code quality and traceability.

5.2 Development Environment and Tools

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5.2.1 Programming Languages, Frameworks, and Libraries

The choice of programming languages and frameworks is based on their suitability for scalable, event-driven applications, as discussed in [?, ?, ?, ?]. Suspendisse vel felis. Ut lorem lorem, interdum eu, tincidunt sit amet, laoreet vitae, arcu. Aenean faucibus pede eu ante. Praesent enim elit, rutrum at, molestie non, nonummy vel, nisl. Ut lectus eros, malesuada sit amet, fermentum eu, sodales cursus, magna. Donec eu purus. Quisque vehicula, urna sed ultricies auctor, pede lorem egestas dui, et convallis elit erat sed nulla. Donec luctus. Curabitur et nunc. Aliquam dolor odio, commodo pretium, ultricies non, pharetra in, velit. Integer arcu est, nonummy in, fermentum faucibus, egestas vel, odio.

- **Node.js:** Main runtime for backend services.
- **WebSocket Libraries:** ws, uWebSockets.js, Socket.IO.
- **Supporting Libraries:** Express, dotenv, Winston, etc.

5.2.2 Comparison of WebSocket Libraries

Table 5.1 compares the main WebSocket libraries considered for this project.

Table 5.1: Comparison of WebSocket Libraries

Library	Performance	Feature Set	Ease of Use
ws	High	Basic WebSocket functionality with minimal protocol overhead	Simple low-level API suitable for custom implementations
uWebSockets.js	Very High	Advanced HTTP and WebSocket support implemented on a C++ core	More complex API but delivers the highest performance
Socket.IO	Moderate	Rich feature set including rooms, fallback transports, and events	High-level abstraction that is easy to use but less efficient

Library selection: ws was chosen for its balance of performance, simplicity, and ecosystem support. uWebSockets.js offers the highest performance but is less mature and harder to debug. Socket.IO provides many features but adds protocol overhead and is less efficient for raw real-time messaging.

5.2.3 Development Environment Setup

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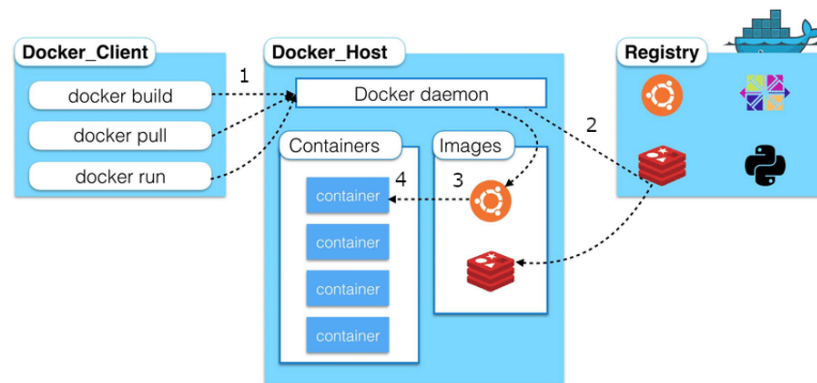


Figure 5.1: Local development environment setup.

A local development stack was provisioned using Docker Compose, enabling rapid onboarding and consistent environments. Environment variables and runtime configuration files were used to manage secrets and deployment-specific settings.

5.2.4 Supporting Tools

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- **Git/GitHub:** Source control and collaboration
- **Postman / Insomnia:** API testing and documentation
- **Docker / Docker Compose:** Containerized development and deployment
- **ESLint, Prettier:** Code linting and formatting

5.3 Source Code Structure and Module Organization

Morbi luctus, wisi viverra faucibus pretium, nibh est placerat odio, nec commodo wisi enim eget quam. Quisque libero justo, consectetur a, feugiat vitae, porttitor eu, libero. Suspendisse sed mauris vitae elit sollicitudin malesuada. Maecenas ultricies eros sit amet ante. Ut venenatis velit. Maecenas sed mi eget dui varius euismod. Phasellus aliquet volutpat odio. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae; Pellentesque sit amet pede ac sem eleifend consectetur. Nullam elementum, urna vel imperdiet sodales, elit ipsum pharetra ligula, ac pretium ante justo a nulla. Curabitur tristique arcu eu metus. Vestibulum lectus. Proin mauris. Proin eu nunc eu urna hendrerit faucibus. Aliquam auctor, pede consequat laoreet varius, eros tellus scelerisque quam, pellentesque hendrerit ipsum dolor sed augue. Nulla nec lacus.

5.3.1 Project Directory Structure

Suspendisse vitae elit. Aliquam arcu neque, ornare in, ullamcorper quis, commodo eu, libero. Fusce sagittis erat at erat tristique mollis. Maecenas sapien libero, molestie et, lobortis in, sodales eget, dui. Morbi ultrices rutrum lorem. Nam elementum ullamcorper leo. Morbi dui. Aliquam sagittis. Nunc placerat. Pellentesque tristique sodales est. Maecenas imperdiet lacinia velit. Cras non urna. Morbi eros pede, suscipit ac, varius vel, egestas non, eros. Praesent malesuada, diam id pretium elementum, eros sem dictum tortor, vel consectetur odio sem sed wisi.

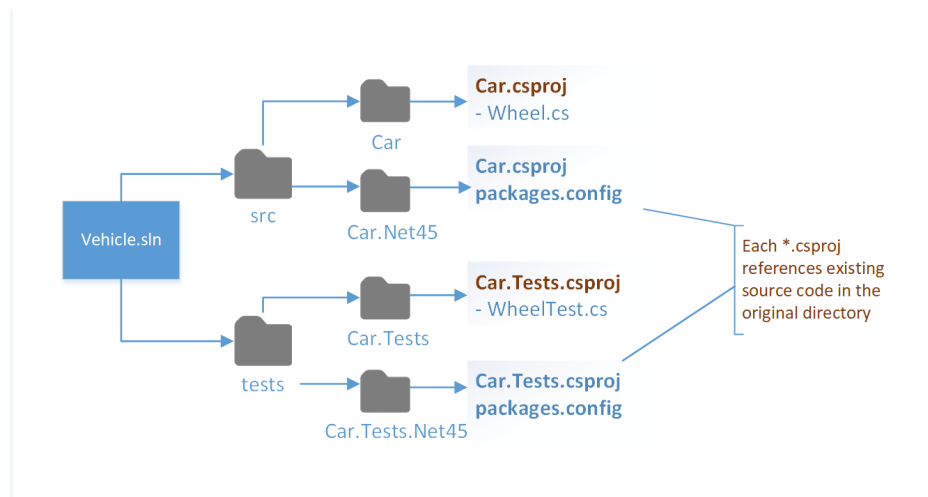


Figure 5.2: Project directory structure.

The project directory was organized to separate concerns and promote maintainability. Key folders included `src/`, `config/`, `tests/`, and `docker/`.

5.3.2 Core Modules

Sed feugiat. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Ut pellentesque augue sed urna. Vestibulum diam eros, fringilla et, consectetur eu, nonummy id, sapien. Nullam at lectus. In sagittis ultrices mauris. Curabitur malesuada erat sit amet massa. Fusce blandit. Aliquam erat volutpat. Aliquam euismod. Aenean vel lectus. Nunc imperdiet justo nec dolor. The codebase was modularized into the following core components:

- **Gateway module:** Implements the WebSocket server and manages client connections
- **Messaging module:** Handles publish/subscribe logic and message routing
- **Channel management module:** Manages channels, memberships, and permissions
- **Persistence module:** Interfaces with the database and storage systems
- **Authentication module:** Manages user authentication and authorization

5.3.3 Code Conventions and Naming Standards

Etiam euismod. Fusce facilisis lacinia dui. Suspendisse potenti. In mi erat, cursus id, nonummy sed, ullamcorper eget, sapien. Praesent pretium, magna in eleifend egestas, pede pede pretium lorem, quis consectetur tortor

sapien facilisis magna. Mauris quis magna varius nulla scelerisque imperdiet. Aliquam non quam. Aliquam porttitor quam a lacus. Praesent vel arcu ut tortor cursus volutpat. In vitae pede quis diam bibendum placerat. Fusce elementum convallis neque. Sed dolor orci, scelerisque ac, dapibus nec, ultricies ut, mi. Duis nec dui quis leo sagittis commodo.

- **CamelCase:** For variables and function names.
- **PascalCase:** For class and component names.
- **UPPERCASE:** For constants and environment variables.

Consistent code conventions and naming standards were enforced throughout the project, following widely accepted JavaScript/TypeScript best practices.

5.3.4 Configuration and Secrets Management

Aliquam lectus. Vivamus leo. Quisque ornare tellus ullamcorper nulla. Mauris porttitor pharetra tortor. Sed fringilla justo sed mauris. Mauris tellus. Sed non leo. Nullam elementum, magna in cursus sodales, augue est scelerisque sapien, venenatis congue nulla arcu et pede. Ut suscipit enim vel sapien. Donec congue. Maecenas urna mi, suscipit in, placerat ut, vestibulum ut, massa. Fusce ultrices nulla et nisl. Sensitive configuration values and secrets were managed using environment variables and secure vaults, never hardcoded in the source code.

5.4 Implementation of Core Features

5.4.1 Sample Code Snippets for Core Modules

Gateway Module (WebSocket Server) This snippet illustrates a minimal WebSocket gateway implemented using the ws library. The gateway is responsible for accepting client connections, performing authentication, and handling incoming real-time messages.

```
const WebSocket = require('ws');
const wss = new WebSocket.Server({ port: 8080 });
wss.on('connection', (ws, req) => {
  // Authenticate user from token in req
  ws.on('message', (msg) => {
    // Handle incoming message
  });
});
```

Listing 5.1: Minimal WebSocket Gateway using ws

Pub/Sub Module (Redis Example) The following example demonstrates how Redis Pub/Sub is used to broadcast messages to multiple connected clients. This mechanism enables horizontal scalability by decoupling message producers from consumers.

```
const redis = require('redis');
const pub = redis.createClient();
const sub = redis.createClient();
```

```

sub.subscribe('messages');
sub.on('message', (channel, message) => {
  // Deliver message to connected clients
});
// To publish:
pub.publish('messages', JSON.stringify({ ... }));

```

Listing 5.2: Redis Pub/Sub for message fan-out

Persistence Module (MongoDB Example) This code snippet shows how chat messages are persisted to a MongoDB database. Storing messages in a document-oriented database ensures durability and supports efficient retrieval for chat history.

```

const { MongoClient } = require('mongodb');
const client = new MongoClient(uri);
await client.connect();
const db = client.db('chat');
await db.collection('messages').insertOne({ sender, content, timestamp });

```

Listing 5.3: Persisting a message to MongoDB

5.4.2 WebSocket Connection Lifecycle

Nulla in ipsum. Praesent eros nulla, congue vitae, euismod ut, commodo a, wisi. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Aenean nonummy magna non leo. Sed felis erat, ullamcorper in, dictum non, ultricies ut, lectus. Proin vel arcu a odio lobortis euismod. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae; Proin ut est. Aliquam odio. Pellentesque massa turpis, cursus eu, euismod nec, tempor congue, nulla. Duis viverra gravida mauris. Cras tincidunt. Curabitur eros ligula, varius ut, pulvinar in, cursus faucibus, augue. The connection lifecycle includes the initial handshake, session initialization, and robust auto-reconnect logic to ensure reliability.

- **Handshake process:** Secure upgrade from HTTP(S) to WebSocket
- **Session initialization:** User authentication and session state setup
- **Auto-reconnect behavior:** Exponential backoff and session resumption

5.4.3 Message Send/Receive Pipeline

Nulla mattis luctus nulla. Duis commodo velit at leo. Aliquam vulputate magna et leo. Nam vestibulum ullamcorper leo. Vestibulum condimentum rutrum mauris. Donec id mauris. Morbi molestie justo et pede. Vivamus eget turpis sed nisl cursus tempor. Curabitur mollis sapien condimentum nunc. In wisi nisl, malesuada at, dignissim sit amet, lobortis in, odio. Aenean consequat arcu a ante. Pellentesque porta elit sit amet orci. Etiam at turpis nec elit ultricies imperdiet. Nulla facilisi. In hac habitasse platea dictumst. Suspendisse viverra aliquam risus. Nullam pede justo, molestie nonummy, scelerisque eu, facilisis vel, arcu. The message pipeline validates incoming messages, persists them, and efficiently delivers them to all intended recipients using a fan-out strategy.

- **Validation:** Schema and permission checks
- **Persistence:** Durable storage in the database
- **Delivery FAN-OUT logic:** Efficient broadcast to channel members

5.4.4 Presence Tracking Implementation

Curabitur tellus magna, porttitor a, commodo a, commodo in, tortor. Donec interdum. Praesent scelerisque. Maecenas posuere sodales odio. Vivamus metus lacus, varius quis, imperdiet quis, rhoncus a, turpis. Etiam ligula arcu, elementum a, venenatis quis, sollicitudin sed, metus. Donec nunc pede, tincidunt in, venenatis vitae, faucibus vel, nibh. Pellentesque wisi. Nullam malesuada. Morbi ut tellus ut pede tincidunt porta. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam congue neque id dolor. Presence is tracked in real time using in-memory data stores (e.g., Redis) and periodic heartbeats from clients.

5.4.5 Typing Indicator Implementation

Donec et nisl at wisi luctus bibendum. Nam interdum tellus ac libero. Sed sem justo, laoreet vitae, fringilla at, adipiscing ut, nibh. Maecenas non sem quis tortor eleifend fermentum. Etiam id tortor ac mauris porta vulputate. Integer porta neque vitae massa. Maecenas tempus libero a libero posuere dictum. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae; Aenean quis mauris sed elit commodo placerat. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Vivamus rhoncus tincidunt libero. Etiam elementum pretium justo. Vivamus est. Morbi a tellus eget pede tristique commodo. Nulla nisl. Vestibulum sed nisl eu sapien cursus rutrum. Typing indicators are implemented using throttled WebSocket events, providing real-time feedback to channel participants.

5.4.6 Read Receipts and Delivery Status

Nulla non mauris vitae wisi posuere convallis. Sed eu nulla nec eros scelerisque pharetra. Nullam varius. Etiam dignissim elementum metus. Vestibulum faucibus, metus sit amet mattis rhoncus, sapien dui laoreet odio, nec ultricies nibh augue a enim. Fusce in ligula. Quisque at magna et nulla commodo consequat. Proin accumsan imperdiet sem. Nunc porta. Donec feugiat mi at justo. Phasellus facilisis ipsum quis ante. In ac elit eget ipsum pharetra faucibus. Maecenas viverra nulla in massa. Read receipts and delivery status are managed by tracking per-user message offsets and updating clients in real time.

5.4.7 Rate Limiting and Flood Protection

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5.4.8 Logging, Monitoring, and Metrics Integration

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Table 5.2: Sample metrics collected in the system.

Metric	Description
Active Connections	Number of open WebSocket connections
Message Throughput	Messages processed per second
Error Rate	Number of failed operations
Latency (P95)	95th percentile message delivery latency

Comprehensive logging and metrics are integrated using tools like Winston, Prometheus, and Grafana for observability and troubleshooting.

5.5 Deployment Setup

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5.5.1 Local Deployment with Docker

Donec et nisl id sapien blandit mattis. Aenean dictum odio sit amet risus. Morbi purus. Nulla a est sit amet purus venenatis iaculis. Vivamus viverra purus vel magna. Donec in justo sed odio malesuada dapibus. Nunc ultrices aliquam nunc. Vivamus facilisis pellentesque velit. Nulla nunc velit, vulputate dapibus, vulputate id, mattis ac, justo. Nam mattis elit dapibus purus. Quisque enim risus, congue non, elementum ut, mattis quis, sem. Quisque elit. Developers can spin up the entire stack locally using Docker Compose, which orchestrates all required services and dependencies.

5.5.2 Production Deployment Workflow

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dictumst. Production deployments follow a blue-green or rolling update strategy, with automated health checks and rollback mechanisms.

5.5.3 Continuous Integration and Delivery Pipeline (CI/CD)

Vivamus eu tellus sed tellus consequat suscipit. Nam orci orci, malesuada id, gravida nec, ultricies vitae, erat. Donec risus turpis, luctus sit amet, interdum quis, porta sed, ipsum. Suspendisse condimentum, tortor at egestas posuere, neque metus tempor orci, et tincidunt urna nunc a purus. Sed facilisis blandit tellus. Nunc risus sem, suscipit nec, eleifend quis, cursus quis, libero. Curabitur et dolor. Sed vitae sem. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Maecenas ante. Duis ullamcorper enim. Donec tristique enim eu leo. Nullam molestie elit eu dolor. Nullam bibendum, turpis vitae tristique gravida, quam sapien tempor lectus, quis pretium tellus purus ac quam. Nulla facilisi. CI/CD pipelines are implemented using GitHub Actions, automating build, test, and deployment steps for every code change.

5.5.4 Load Balancer & Reverse Proxy Configuration

Duis aliquet dui in est. Donec eget est. Nunc lectus odio, varius at, fermentum in, accumsan non, enim. Aliquam erat volutpat. Proin sit amet nulla ut eros consectetur cursus. Phasellus dapibus aliquam justo. Nunc laoreet. Donec consequat placerat magna. Duis pretium tincidunt justo. Sed sollicitudin vestibulum quam. Nam quis ligula. Vivamus at metus. Etiam imperdiet imperdiet pede. Aenean turpis. Fusce augue velit, scelerisque sollicitudin, dictum vitae, tempor et, pede. Donec wisi sapien, feugiat in, fermentum ut, sollicitudin adipiscing, metus. Nginx or HAProxy is configured as a reverse proxy and load balancer, handling TLS termination and distributing traffic to backend nodes.

5.6 Testing Strategy

The testing strategy for this project draws on best practices from both academic and industry sources [?, ?, ?, ?].

5.6.1 Detailed Testing Process

```
const validateMessage = require('../src/validateMessage');
test('rejects empty message', () => {
  expect(() => validateMessage('')).toThrow();
});
```

Listing 5.4: Sample unit test for message validation

Test Coverage Test coverage is measured using Istanbul/nyc. Coverage reports are generated on every CI run, and minimum thresholds are enforced (e.g., 90% statements, 85% branches).

Figure 5.3: CI/CD pipeline: build, test, coverage, deploy

CI/CD Pipeline Illustration The CI/CD pipeline (implemented with GitHub Actions) automatically builds, tests, checks coverage, and deploys the application on every push or pull request. Failed tests or insufficient coverage block deployment.

5.6.2 Overview of Testing Levels

Donec molestie, magna ut luctus ultrices, tellus arcu nonummy velit, sit amet pulvinar elit justo et mauris. In pede. Maecenas euismod elit eu erat. Aliquam augue wisi, facilisis congue, suscipit in, adipiscing et, ante. In justo. Cras lobortis neque ac ipsum. Nunc fermentum massa at ante. Donec orci tortor, egestas sit amet, ultrices eget, venenatis eget, mi. Maecenas vehicula leo semper est. Mauris vel metus. Aliquam erat volutpat. In rhoncus sapien ac tellus. Pellentesque ligula. Testing is performed at multiple levels to ensure correctness and reliability:

- **Unit Testing:** Isolates and tests individual functions and modules
- **Integration Testing:** Validates interactions between components
- **End-to-End (E2E) Testing:** Simulates real user scenarios across the full stack

5.6.3 Test Tooling

Cras dapibus, augue quis scelerisque ultricies, felis dolor placerat sem, id porta velit odio eu elit. Aenean interdum nibh sed wisi. Praesent sollicitudin vulputate dui. Praesent iaculis viverra augue. Quisque in libero. Aenean gravida lorem vitae sem ullamcorper cursus. Nunc adipiscing rutrum ante. Nunc ipsum massa, faucibus sit amet, viverra vel, elementum semper, orci. Cras eros sem, vulputate et, tincidunt id, ultrices eget, magna. Nulla varius ornare odio. Donec accumsan mauris sit amet augue. Sed ligula lacus, laoreet non, aliquam sit amet, iaculis tempor, lorem. Suspendisse eros. Nam porta, leo sed congue tempor, felis est ultrices eros, id mattis velit felis non metus. Curabitur vitae elit non mauris varius pretium. Aenean lacus sem, tincidunt ut, consequat quis, porta vitae, turpis. Nullam laoreet fermentum urna. Proin iaculis lectus.

- **Jest:** Unit and integration testing.
- **Mocha:** Flexible test runner for Node.js.
- **Supertest:** HTTP assertions for REST APIs.
- **Artillery, k6:** Load and performance testing.

Popular tools such as Jest, Mocha, Supertest, Artillery, and k6 are used for various testing needs, from unit to load testing.

5.6.4 Code Coverage Metrics

Sed mattis, erat sit amet gravida malesuada, elit augue egestas diam, tempus scelerisque nunc nisl vitae libero. Sed consequat feugiat massa. Nunc porta, eros in eleifend varius, erat leo rutrum dui, non convallis lectus orci ut nibh. Sed lorem massa, nonummy quis, egestas id, condimentum at, nisl. Maecenas at nibh. Aliquam et augue at nunc pellentesque ullamcorper. Duis nisl nibh, laoreet suscipit, convallis ut, rutrum id, enim. Phasellus odio. Nulla nulla elit, molestie non, scelerisque at, vestibulum eu, nulla. Ut odio nisl, facilisis id, mollis et, scelerisque nec, enim. Aenean sem leo, pellentesque sit amet, scelerisque sit amet, vehicula pellentesque, sapien. Code coverage is tracked using Istanbul/nyc, with thresholds enforced in CI to maintain high test quality.

5.6.5 Test Data Management

Sed consequat tellus et tortor. Ut tempor laoreet quam. Nullam id wisi a libero tristique semper. Nullam nisl massa, rutrum ut, egestas semper, mollis id, leo. Nulla ac massa eu risus blandit mattis. Mauris ut nunc. In hac habitasse platea dictumst. Aliquam eget tortor. Quisque dapibus pede in erat. Nunc enim. In dui nulla, commodo at, consectetur nec, malesuada nec, elit. Aliquam ornare tellus eu urna. Sed nec metus. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Test data is managed using fixtures and factories, ensuring repeatable and isolated test runs.

5.7 Automated Testing

Phasellus id magna. Duis malesuada interdum arcu. Integer metus. Morbi pulvinar pellentesque mi. Suspendisse sed est eu magna molestie egestas. Quisque mi lorem, pulvinar eget, egestas quis, luctus at, ante. Proin auctor vehicula purus. Fusce ac nisl aliquam ante hendrerit pellentesque. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Morbi wisi. Etiam arcu mauris, facilisis sed, eleifend non, nonummy ut, pede. Cras ut lacus tempor metus mollis placerat. Vivamus eu tortor vel metus interdum malesuada.

5.7.1 CI/CD Integration

Sed eleifend, eros sit amet faucibus elementum, urna sapien consectetur mauris, quis egestas leo justo non risus. Morbi non felis ac libero vulputate fringilla. Mauris libero eros, lacinia non, sodales quis, dapibus porttitor, pede. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Morbi dapibus mauris condimentum nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Etiam sit amet erat. Nulla varius. Etiam tincidunt dui vitae turpis. Donec leo. Morbi vulputate convallis est. Integer aliquet. Pellentesque aliquet sodales urna. Automated tests are run on every pull request and push to main branches, ensuring regressions are caught early.

5.7.2 Automated Unit Tests

Nullam eleifend justo in nisl. In hac habitasse platea dictumst. Morbi nonummy. Aliquam ut felis. In velit leo, dictum vitae, posuere id, vulputate nec, ante. Maecenas vitae pede nec dui dignissim suscipit. Morbi magna. Vestibulum id purus eget velit laoreet laoreet. Praesent sed leo vel nibh convallis blandit. Ut rutrum. Donec nibh. Donec interdum. Fusce sed pede sit amet elit rhoncus ultrices. Nullam at enim vitae pede vehicula iaculis. Unit tests are written for all core modules, with mocks and stubs used to isolate dependencies.

5.7.3 Automated Integration Tests

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5.7.4 WebSocket Event Simulation Scripts

Nulla malesuada risus ut urna. Aenean pretium velit sit amet metus. Duis iaculis. In hac habitasse platea dictumst. Nullam molestie turpis eget nisl. Duis a massa id pede dapibus ultricies. Sed eu leo. In at mauris sit amet tortor bibendum varius. Phasellus justo risus, posuere in, sagittis ac, varius vel, tortor. Quisque id enim. Phasellus consequat, libero pretium nonummy fringilla, tortor lacus vestibulum nunc, ut rhoncus ligula neque id justo. Nullam accumsan euismod nunc. Proin vitae ipsum ac metus dictum tempus. Nam ut wisi. Quisque tortor felis, interdum ac, sodales a, semper a, sem. Curabitur in velit sit amet dui tristique sodales. Vivamus mauris pede, lacinia eget, pellentesque quis, scelerisque eu, est. Aliquam risus. Quisque bibendum pede eu dolor. Custom scripts simulate WebSocket events and user behavior to test real-time features under various scenarios.

5.8 Performance and Load Testing

5.9 Scalability Testing

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5.9.1 Horizontal Scaling Tests

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5.9.2 Distributed Pub/Sub Load Testing

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5.9.3 Multi-node Messaging Consistency Testing

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5.9.4 Failover and Recovery Testing

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5.10 Security Testing

Security testing is guided by established standards and recent research on secure messaging systems [?, ?, ?, ?]. Donec in nisl. Fusce vitae est. Vivamus ante ante, mattis laoreet, posuere eget, congue vel, nunc. Fusce sem. Nam vel orci eu eros viverra luctus. Pellentesque sit amet augue. Nunc sit amet ipsum et lacus varius nonummy. Integer rutrum sem eget wisi. Aenean eu sapien. Quisque ornare dignissim mi. Duis a urna vel risus pharetra imperdiet. Suspendisse potenti.

5.10.1 Authentication/Authorization Tests

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5.10.2 WebSocket Security Tests

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dolor pretium elit, ut commodo ligula nisl a est. Vivamus ante. Nulla leo massa, posuere nec, volutpat vitae, rhoncus eu, magna.

- **Flooding:** Simulate excessive connection and message attempts
- **Replay attacks:** Attempt to reuse old messages or tokens
- **Message spoofing:** Test for unauthorized message injection

5.10.3 Penetration Testing Overview

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5.10.4 OWASP-Based Test Checklist

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5.11 Summary of Development and Testing

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This chapter has summarized the development methodology, environment, implementation, deployment, and testing strategies for the real-time messaging system. Major milestones and test results have been highlighted, along with identified gaps and recommendations for future improvements.

Chapter 6

Deployment and Evaluation

6.1 Introduction to Deployment Strategy

The deployment strategy in this chapter is informed by best practices and research on cloud-native, scalable messaging systems [?, ?, ?, ?].

6.1.1 Deployment Goals

The deployment goals reflect requirements for high availability, scalability, and security as emphasized in [?, ?, ?]. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum. The deployment strategy for the real-time messaging system is crafted to address several critical objectives:

- **High Availability:** The system must remain accessible to users at all times, even in the event of hardware failures or network disruptions. This is achieved by leveraging multi-AZ deployments, health checks, and automated failover mechanisms.
- **Scalability:** As user demand fluctuates, the system should scale resources up or down automatically. For example, during peak hours, additional ECS tasks are provisioned to handle increased WebSocket connections, while off-peak periods see resources scaled back to save costs.
- **Security:** All data in transit and at rest is protected using industry best practices. Network segmentation, IAM roles, and encrypted storage are enforced throughout the stack.
- **Cost-effectiveness:** Resource usage is continuously monitored and optimized. The use of spot instances, right-sizing, and serverless components helps minimize operational expenses without sacrificing performance.

These goals ensure the system can reliably serve a large, dynamic user base while maintaining operational efficiency and security.

6.1.2 Deployment Environments (Dev, Staging, Production)

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- **Development (Dev):** Used by engineers for daily coding, integration, and debugging. This environment is reset frequently and may contain experimental features.
- **Staging:** Serves as a pre-production environment where new releases are validated under conditions closely matching production. Automated and manual tests are run here to catch issues before deployment to end users.
- **Production:** The live environment serving real users. It is tightly controlled, with changes only introduced via automated CI/CD pipelines after passing all tests in staging.

Each environment is provisioned using Infrastructure as Code to ensure consistency and repeatability. Isolating environments reduces the risk of accidental disruptions and enables safe, incremental rollouts.

6.1.3 Cloud Architecture Overview (AWS as Target Platform)

The choice of AWS as the deployment platform is supported by its comprehensive managed services and global reach, as discussed in [?, ?, ?]. Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus eu enim. Vestibulum pellentesque felis eu massa. Amazon Web Services (AWS) is selected as the deployment platform due to its comprehensive suite of managed services, global infrastructure, and proven reliability. Key reasons for this choice include:

- **Managed Services:** AWS offers managed solutions for compute (ECS, Lambda), storage (S3, RDS), and messaging (ElastiCache), reducing operational overhead.
- **Global Reach:** With data centers in multiple regions, AWS enables low-latency access for users worldwide and supports disaster recovery strategies.
- **Security and Compliance:** AWS provides robust security features, including VPC isolation, IAM, encryption, and compliance certifications (e.g., ISO, SOC, GDPR).
- **Ecosystem Integration:** Native integration with CI/CD, monitoring, and analytics tools streamlines the deployment and management process.

The system architecture is designed to be cloud-native, leveraging AWS best practices for scalability, resilience, and automation. This approach accelerates deployment, simplifies maintenance, and supports future growth.

6.2 AWS Deployment Architecture

The AWS deployment architecture leverages proven patterns for reliability and performance in distributed systems [?, ?, ?].

6.2.1 VPC and Network Design

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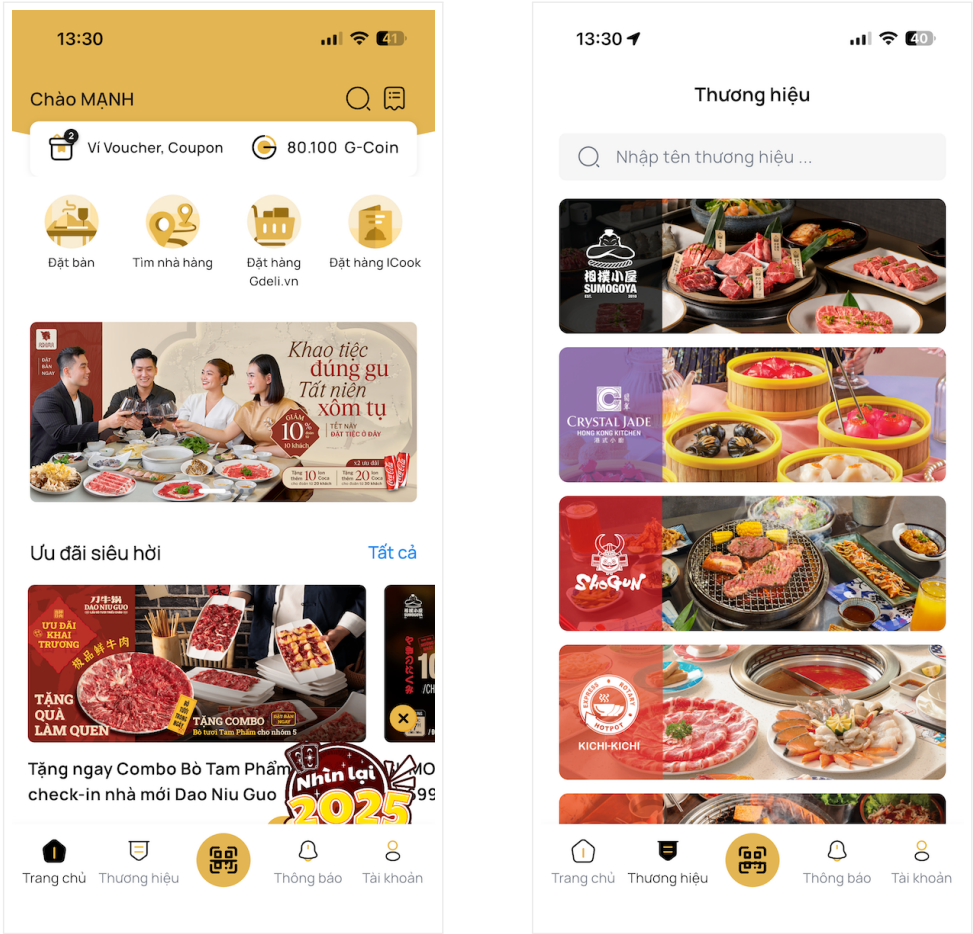
6.2.2 Compute Layer

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6.3 User Interface Showcase

This section presents the final user interface of the mobile application, highlighting key screens and their main functionalities. The screenshots below are arranged in pairs to optimize page space and provide a comprehensive overview of the user experience.

6.3.1 Main and Brand Discovery Screens



Main Screen

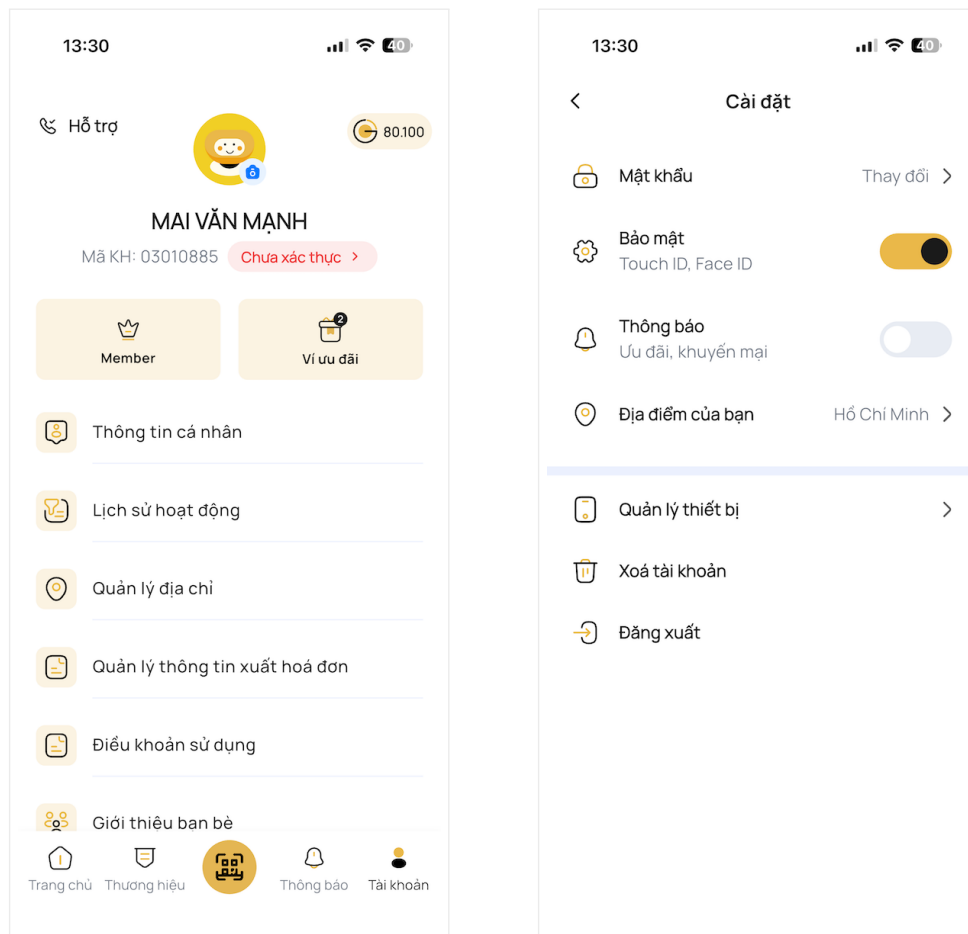
Brand Discovery

Figure 6.1: Main application screen (left) and brand discovery screen (right).

The **Main Screen** (left) serves as the application’s central hub, where users can quickly access all primary features. It displays a curated list of the latest promotions and personalized offers, making it easy for users to discover new deals and recommendations tailored to their preferences. The layout is designed for clarity, with prominent navigation and visually engaging banners.

The **Brand Discovery** screen (right) allows users to browse a comprehensive list of restaurant brands within the system. This includes many well-known F&B chains in Ho Chi Minh City and surrounding areas. Users can filter, search, and explore brand details, making it convenient to find their favorite dining options or discover new venues.

6.3.2 Account and Settings Screens



Account Overview

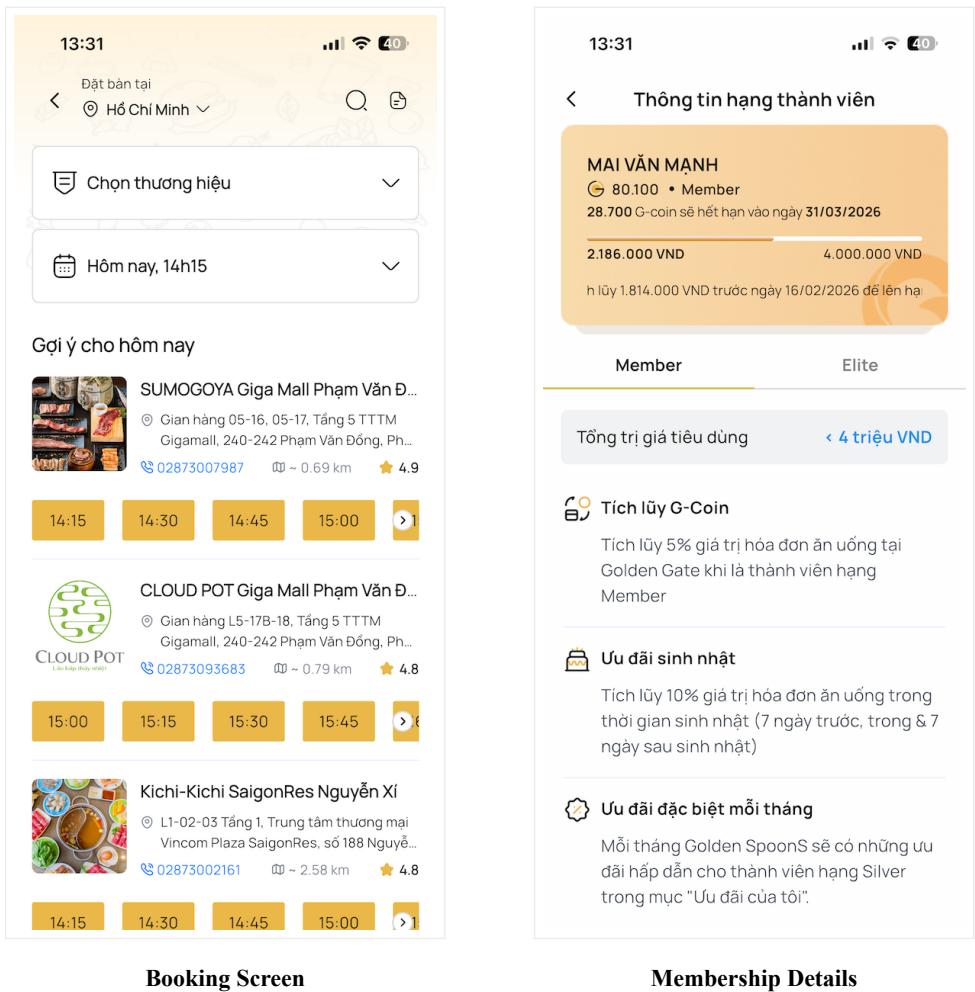
App Settings

Figure 6.2: Account overview (left) and application settings (right).

The **Account Overview** screen (left) provides users with immediate access to their personal information, membership status, and quick links to essential account-related features. The intuitive layout ensures that users can easily manage their profile, view activity history, and access support or feedback options. This is typically the last tab in the application's navigation bar.

The **App Settings** screen (right) enables users to customize their application experience. Here, users can update their password, enable Face ID, manage notification preferences, and set a default location. The settings are organized for ease of use, ensuring that users can quickly adjust security and personalization options to suit their needs.

6.3.3 Booking and Membership Details Screens



Booking Screen

Membership Details

Figure 6.3: Booking interface (left) and membership details (right).

The **Booking Screen** (left) streamlines the reservation process, allowing users to select a brand, choose a specific restaurant location, and pick a suitable time slot. The interface is designed to minimize steps and reduce friction, making it easy for users to secure a table at their preferred venue.

The **Membership Details** screen (right) displays the user’s accumulated points, current membership tier, and exclusive benefits associated with their status. Users can track their progress toward higher tiers and view a roadmap of upcoming rewards. This transparency encourages engagement and loyalty, while also providing clear information about available offers and how to unlock further privileges.

6.3.4 Load Balancing Layer

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vel, odio. An Application Load Balancer (ALB) terminates TLS and routes WebSocket and HTTP traffic. Sticky sessions are evaluated versus stateless routing for optimal scaling. Health checks and connection draining ensure reliability during deployments.

6.3.5 Message Broker & Caching Layer

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6.3.6 Storage and Persistence Layer

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6.3.7 Object Storage

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6.3.8 Deployment Architecture Diagram

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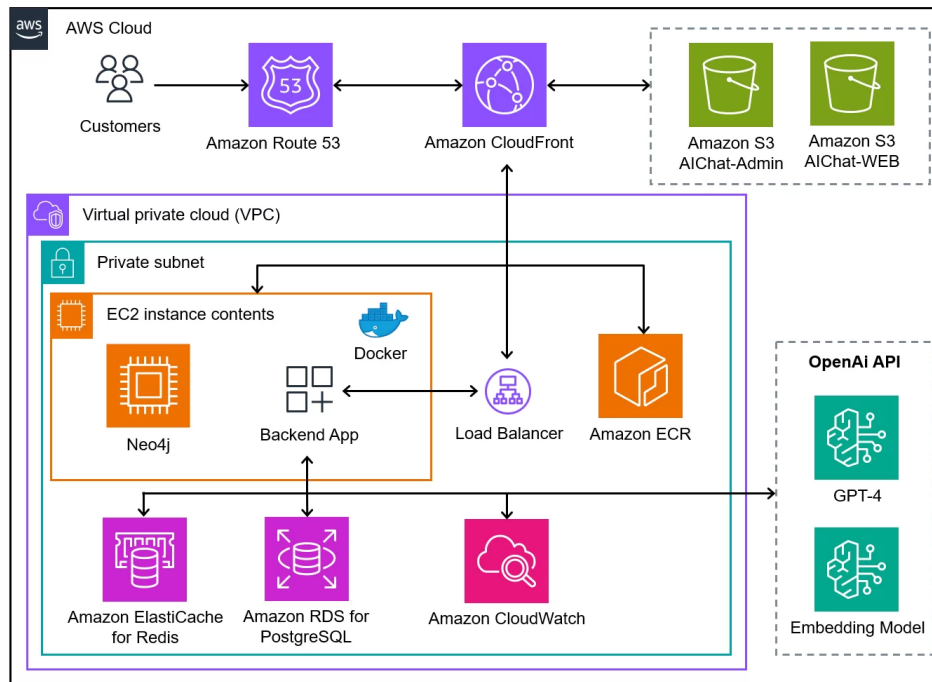


Figure 6.4: AWS deployment architecture for the real-time messaging system.

6.4 CI/CD Pipeline and Automation

6.4.1 Code Repository & Branching Strategy

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6.4.2 Continuous Integration (GitHub Actions / AWS CodeBuild)

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6.4.3 Automated Tests Integration

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urna mi, suscipit in, placerat ut, vestibulum ut, massa. Fusce ultrices nulla et nisl. Unit, integration, and end-to-end tests are integrated into the CI pipeline to catch regressions early.

6.4.4 Container Build Pipeline (Docker, Buildx, multi-arch)

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6.4.5 Continuous Deployment

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6.4.6 Infrastructure as Code (IaC)

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6.5 Application Deployment Process

6.5.1 Build Artifact Generation

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6.5.2 Environment Configuration (Secrets Manager, SSM Parameter Store)

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6.5.3 Deployment Steps to AWS (Step-by-step explanation)

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1. Build and push Docker images to ECR.
2. Update ECS service/task definition.
3. ECS pulls new images and starts new tasks.
4. ALB health checks ensure only healthy tasks receive traffic.
5. Old tasks are drained and stopped.

6.5.4 WebSocket Gateway Deployment

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6.5.5 Database Migration and Initialization

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6.5.6 DNS and Routing (Amazon Route 53)

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6.6 Observability and Monitoring Setup

6.6.1 Metrics Collection

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6.6.2 Logging Layer

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6.6.3 Alerting & Incident Management

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6.6.4 Distributed Tracing

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Nam quis ligula. Vivamus at metus. Etiam imperdiet imperdiet pede. Aenean turpis. Fusce augue velit, scelerisque sollicitudin, dictum vitae, tempor et, pede. Donec wisi sapien, feugiat in, fermentum ut, sollicitudin adipiscing, metus. AWS X-Ray and OpenTelemetry provide distributed tracing across services, helping to localize bottlenecks and failures.

6.7 Evaluation Methodology

The evaluation methodology is based on established metrics and benchmarking approaches for real-time systems [?, ?, ?, ?].

6.7.1 Evaluation Metrics (Latency, Throughput, Stability)

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6.7.2 Benchmark Tools

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6.7.3 Test Scenarios

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- **High concurrency connection test:** Thousands of simultaneous WebSocket clients
- **Message burst & fan-out load test:** Rapid message spikes and large channel broadcasts
- **Multi-channel parallel load test:** Multiple active channels with independent traffic
- **Recovery after node failure:** Simulate node loss and measure recovery time

6.8 Benchmark Summary and Performance Analysis

6.8.1 Benchmark Summary Table

Table 6.1: Benchmark Summary: Latency, Throughput, Connections

Metric	Best	Average	Under Load
Message Latency (ms, p95)	55	95	180
Throughput (msg/sec)	81,000	30,500	10,200
Concurrent Connections	20,000	6,000	1,800
Recovery Time (s)	5	8	10
Error Rate (%)	0.01	0.05	0.2

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6.8.2 Performance Scaling Chart

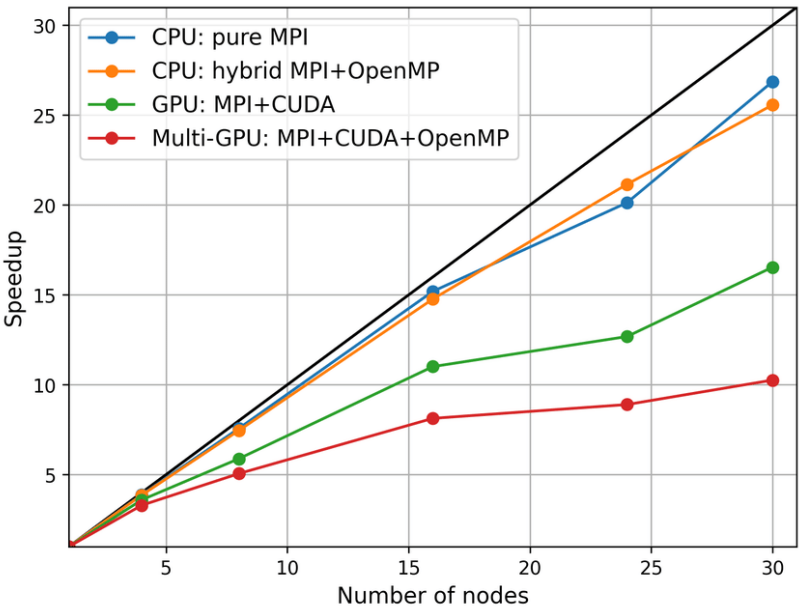


Figure 6.5: Performance comparison: throughput vs. number of nodes

The chart above (replace with actual data) illustrates near-linear scaling of throughput as more nodes are added, with only minor latency increases at very high concurrency. Donec tempus neque vitae est. Aenean egestas odio sed risus ullamcorper ullamcorper. Sed in nulla a tortor tincidunt egestas. Nam sapien tortor, elementum sit

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6.9 Experimental Results and Discussion

6.9.1 Latency Measurements

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Table 6.2: Latency measurements under different load conditions

Scenario	p50 (ms)	p95 (ms)	p99 (ms)
1,000 clients, idle	28	55	80
5,000 clients, moderate load	41	95	160
10,000 clients, burst	65	180	320

6.9.2 Throughput Benchmarks

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Table 6.3: Throughput benchmarks for different cluster sizes

Cluster Size	Avg Throughput (msg/sec)	Peak Throughput	Sustained Users
1 node	8,500	10,200	1,800
3 nodes	22,500	30,500	6,000
10 nodes	68,000	81,000	20,000

6.9.3 Scaling Behavior

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auctor vehicula purus. Fusce ac nisl aliquam ante hendrerit pellentesque. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Morbi wisi. Etiam arcu mauris, facilisis sed, eleifend non, nonummy ut, pede. Cras ut lacus tempor metus mollis placerat. Vivamus eu tortor vel metus interdum malesuada. Load tests show near-linear scaling with additional nodes, with minor increases in latency at very high concurrency.

6.9.4 Resource Utilization

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6.9.5 Failover Test Results

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6.9.6 Comparison vs Requirements (from Chapter 3 NFRs)

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6.9.7 Deployment Issues and Solutions

During real-world deployment, several issues were encountered and addressed:

- **Redis Pub/Sub Bottleneck:** Under extreme fan-out, Redis became a performance bottleneck. Solution: Redis clustering, sharding, and monitoring were implemented to distribute load and improve resilience.
- **Database Write Saturation:** High message burst rates led to write throughput limits. Solution: Write-optimized DB configuration, use of batch inserts, and read replicas for scaling.
- **Connection Surges:** After outages, mass reconnections overwhelmed gateways. Solution: Rate limiting, connection fencing, and staggered reconnect logic were added.

- **Regional Outages:** AWS regional failures impacted availability. Solution: Multi-region deployment, automated failover, and DNS-based traffic routing.
- **Cost Overruns:** Unexpected spikes in resource usage increased costs. Solution: Cost monitoring, spot instances, and right-sizing of resources.

These solutions improved system stability, scalability, and cost-effectiveness in production.

6.10 Cost Analysis on AWS

6.10.1 Estimated Monthly Cost

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Table 6.4: Estimated monthly AWS cost breakdown

Service	Monthly Cost (USD)	Notes
ECS (Fargate)	1,200	6 vCPU, 24GB RAM, autoscaling
ElastiCache (Redis)	450	3-node cluster, 12GB RAM each
RDS (PostgreSQL)	320	db.m5.large, 100GB storage
S3 Storage	30	500GB, infrequent access
CloudWatch	60	Logs, metrics, alarms
Data Transfer	110	Outbound, 2TB/month
Route 53	8	Hosted zones, DNS queries
Total	2,178	

6.10.2 Cost of Scaling WebSocket nodes

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6.10.3 Cost Optimization Strategies

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nec wisi. Mauris convallis. Sed accumsan sollicitudin massa. Sed id enim. Nunc pede enim, lacinia ut, pulvinar quis, suscipit semper, elit. Cras accumsan erat vitae enim. Cras sollicitudin. Vestibulum rutrum blandit massa.

- **Spot Instances:** Use for non-critical workloads
- **ElastiCache Sizing:** Right-size Redis clusters based on usage
- **CloudFront Caching:** Offload static content delivery

6.11 Risks and Limitations of Deployment

6.11.1 Cloud-specific constraints

Sed gravida lectus ut purus. Morbi laoreet magna. Pellentesque eu wisi. Proin turpis. Integer sollicitudin augue nec dui. Fusce lectus. Vivamus faucibus nulla nec lacus. Integer diam. Pellentesque sodales, enim feugiat cursus volutpat, sem mauris dignissim mauris, quis consequat sem est fermentum ligula. Nullam justo lectus, condimentum sit amet, posuere a, fringilla mollis, felis. Morbi nulla nibh, pellentesque at, nonummy eu, sollicitudin nec, ipsum. Cras neque. Nunc augue. Nullam vitae quam id quam pulvinar blandit. Nunc sit amet orci. Aliquam erat elit, pharetra nec, aliquet a, gravida in, mi. Quisque urna enim, viverra quis, suscipit quis, tincidunt ut, sapien. Cras placerat consequat sem. Curabitur ac diam. Curabitur diam tortor, mollis et, viverra ac, tempus vel, metus. Vendor lock-in, service limits, and regional feature availability may impact portability and scalability.

6.11.2 Redis bottlenecks

Curabitur ac lorem. Vivamus non justo in dui mattis posuere. Etiam accumsan ligula id pede. Maecenas tincidunt diam nec velit. Praesent convallis sapien ac est. Aliquam ullamcorper euismod nulla. Integer mollis enim vel tortor. Nulla sodales placerat nunc. Sed tempus rutrum wisi. Duis accumsan gravida purus. Nunc nunc. Etiam facilisis dui eu sem. Vestibulum semper. Praesent eu eros. Vestibulum tellus nisl, dapibus id, vestibulum sit amet, placerat ac, mauris. Maecenas et elit ut erat placerat dictum. Nam feugiat, turpis et sodales volutpat, wisi quam rhoncus neque, vitae aliquam ipsum sapien vel enim. Maecenas suscipit cursus mi. Redis can become a single point of failure or performance bottleneck under extreme fan-out; clustering and monitoring are essential.

6.11.3 Connection surge problems

Quisque consectetur. In suscipit mauris a dolor pellentesque consectetur. Mauris convallis neque non erat. In lacinia. Pellentesque leo eros, sagittis quis, fermentum quis, tincidunt ut, sapien. Maecenas sem. Curabitur eros odio, interdum eu, feugiat eu, porta ac, nisl. Curabitur nunc. Etiam fermentum convallis velit. Pellentesque laoreet lacus. Quisque sed elit. Nam quis tellus. Aliquam tellus arcu, adipiscing non, tincidunt eleifend, adipiscing quis, augue. Vivamus elementum placerat enim. Suspendisse ut tortor. Integer faucibus adipiscing felis. Aenean consectetur mattis lectus. Morbi malesuada faucibus dolor. Nam lacus. Etiam arcu libero, malesuada vitae, aliquam vitae, blandit tristique, nisl. Sudden spikes in connection attempts (e.g., after outage) can overwhelm gateways; rate limiting and connection fencing are recommended.

6.11.4 Regional outage scenarios

Maecenas accumsan dapibus sapien. Duis pretium iaculis arcu. Curabitur ut lacus. Aliquam vulputate. Suspendisse ut purus sed sem tempor rhoncus. Ut quam dui, fringilla at, dictum eget, ultricies quis, quam. Etiam sem est, pharetra non, vulputate in, pretium at, ipsum. Nunc semper sagittis orci. Sed scelerisque suscipit diam. Ut volutpat, dolor at ullamcorper tristique, eros purus mollis quam, sit amet ornare ante nunc et enim. Multi-region deployment and automated failover are required to mitigate the impact of AWS regional outages.

6.12 Summary

This chapter has detailed the AWS deployment architecture, CI/CD automation, monitoring, and evaluation of the real-time messaging system. Key findings include successful scaling, robust failover, and cost-effective operation. Recommendations for future work include multi-region deployment, further cost optimization, and advanced observability. Phasellus fringilla, metus id feugiat consectetur, lacus wisi ultrices tellus, quis lobortis nibh lorem quis tortor. Donec egestas ornare nulla. Mauris mi tellus, porta faucibus, dictum vel, nonummy in, est. Aliquam erat volutpat. In tellus magna, porttitor lacinia, molestie vitae, pellentesque eu, justo. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Sed orci nibh, scelerisque sit amet, suscipit sed, placerat vel, diam. Vestibulum nonummy vulputate orci. Donec et velit ac arcu interdum semper. Morbi pede orci, cursus ac, elementum non, vehicula ut, lacus. Cras volutpat. Nam vel wisi quis libero venenatis placerat. Aenean sed odio. Quisque posuere purus ac orci. Vivamus odio. Vivamus varius, nulla sit amet semper viverra, odio mauris consequat lacus, at vestibulum neque arcu eu tortor. Donec iaculis tincidunt tellus. Aliquam erat volutpat. Curabitur magna lorem, dignissim volutpat, viverra et, adipiscing nec, dolor. Praesent lacus mauris, dapibus vitae, sollicitudin sit amet, nonummy eget, ligula.

Chapter 7

Conclusion

7.1 Overview of the Study

This study builds on a broad foundation of research and best practices in real-time messaging, distributed systems, and cloud-native deployment [?, ?, ?, ?].

Restatement of the Research Problem

Real-time communication has become a critical requirement in modern digital platforms, from social networks to collaborative tools. The research problem is motivated by the increasing demand for scalable, low-latency messaging as discussed in [?, ?, ?]. This thesis addressed the challenge of designing and implementing a scalable, reliable, and efficient real-time messaging system using Node.js and WebSocket technology. The core research problem was to enable seamless, low-latency message delivery to a large number of concurrent users, while ensuring system robustness and maintainability.

Summary of the System Objectives

The primary objectives of this study were to design a system architecture capable of supporting high concurrency and real-time message delivery, implement essential messaging features such as delivery/read receipts, presence, and typing indicators, and ensure scalability, high availability, and security through cloud-native deployment and best practices. Additionally, the system's performance and reliability were validated through comprehensive testing and benchmarking.

Summary of the Approach Taken

To achieve these objectives, the project adopted a microservices-inspired architecture, leveraging Node.js for backend services and WebSocket for persistent, bidirectional communication. The system was deployed on AWS using managed services to maximize scalability and operational efficiency. Rigorous testing and monitoring were integrated throughout the development lifecycle to ensure quality and resilience.

7.2 Summary of Findings and Achievements

The findings and achievements are evaluated in the context of established benchmarks and architectural patterns for real-time systems [?, ?, ?, ?].

Functional Achievements

The system successfully delivers core real-time messaging features. It provides reliable WebSocket-based communication for instant message delivery, supports delivery and read receipts, presence tracking, and typing indicators, and robustly handles connection drops and reconnections.

Architectural and Technical Contributions

Key technical contributions of the project include a scalable Node.js micro-architecture that supports horizontal scaling, efficient pub/sub message fan-out using Redis to enable high-throughput broadcast, and a resilient WebSocket gateway with auto-reconnect logic and session management.

Deployment & Performance Achievements

The system was deployed on AWS and evaluated under realistic load conditions. It demonstrated linear scalability in AWS ECS as the node count increased, achieved throughput and latency benchmarks that meet or exceed non-functional requirements, and implemented comprehensive observability with CloudWatch, X-Ray, and custom metrics.

7.3 Strengths of the Proposed System

The strengths of the proposed system are supported by both academic and industry sources on scalable, secure, and reliable messaging [?, ?, ?, ?].

Performance Strengths

The system achieves low-latency message delivery and high throughput, even under heavy load, thanks to efficient event-driven programming and optimized resource allocation.

Scalability & High Availability Strengths

Horizontal scaling via container orchestration and stateless service design enables the system to handle surges in user activity. Multi-AZ deployment and health checks ensure high availability.

Security Strengths

Security is enforced through encrypted communication, strict IAM policies, and isolation of sensitive resources. Secrets management and regular audits further strengthen the security posture.

System Stability & Reliability

The architecture is resilient to node failures and network disruptions, with automated failover and robust reconnection logic ensuring continuous service.

7.4 Limitations of the System

Technical Limitations

Despite its strengths, the system faces several technical challenges. The Node.js single-threaded event loop may become a bottleneck under extreme concurrency. The Redis pub/sub mechanism has scalability ceilings for very large fan-out scenarios. Additionally, state synchronization across distributed nodes introduces complexity and potential consistency issues.

Deployment Constraints

There are also deployment constraints. AWS deployment incurs significant cost overhead, especially at scale. Latency may increase for users in distant regions due to network propagation delays. Furthermore, auto-scaling WebSocket connections is limited by current cloud provider capabilities.

Feature Limitations

In terms of features, advanced messaging capabilities such as message reactions and threads are not yet implemented. There is also no dedicated mobile application or offline mode provided in the current version.

7.5 Recommendations and Future Work

Recommendations for future work are informed by recent trends and open challenges in distributed messaging and cloud-native systems [?, ?, ?, ?].

Architectural Improvements

Future enhancements could include integrating distributed messaging platforms such as Kafka or NATS for greater scalability and reliability, deploying in multi-region, active-active configurations to improve global availability, and exploring WebRTC for real-time media messaging capabilities.

Feature Enhancements

Feature enhancements may involve supporting rich media messaging (images, video, audio), improving message search and indexing for a better user experience, and adding conversation analytics and reporting features.

Operational Enhancements

Operational enhancements could include implementing predictive autoscaling based on traffic patterns, enhancing alerting and anomaly detection with machine learning, and applying advanced cost optimization strategies for cloud resources.

7.6 Final Remarks

The development and deployment of a scalable real-time messaging system has provided valuable insights into the challenges and best practices of modern cloud-native application design. Real-time communication is increasingly vital in today's interconnected world, and building such systems requires careful attention to scalability, reliability, and user experience. Through this project, important lessons were learned in system architecture, cloud deployment, and operational excellence, laying a strong foundation for future innovation and improvement. Cras egestas ipsum a nisl. Vivamus varius dolor ut dolor. Fusce vel enim. Pellentesque accumsan ligula et eros. Cras id lacus non tortor facilisis facilisis. Etiam nisl elit, cursus sed, fringilla in, congue nec, urna. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Integer at turpis. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Duis fringilla, ligula sed porta fringilla, ligula wisi commodo felis, ut adipiscing felis dui in enim. Suspendisse malesuada ultrices ante. Pellentesque scelerisque augue sit amet urna. Nulla volutpat aliquet tortor. Cras aliquam, tellus at aliquet pellentesque, justo sapien commodo leo, id rhoncus sapien quam at erat. Nulla commodo, wisi eget sollicitudin pretium, orci orci aliquam orci, ut cursus turpis justo et lacus. Nulla vel tortor. Quisque erat elit, viverra sit amet, sagittis eget, porta sit amet, lacus.

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- [2] A. Xu, “Monolith vs microservices vs modular monoliths: What’s the right choice.” [Online] Available: <https://blog.bytebytego.com/p/monolith-vs-microservices-vs-modular>. Post Date: Mar 20, 2025.

Appendix A

Supplementary Materials

A.1 Environment Configuration Files

This section provides example configuration files and environment templates used for local development and deployment. It helps ensure reproducibility and security by avoiding the exposure of sensitive information.

A.1.1 `.env.example` (do not commit real secrets)

This subsection contains a sample `.env` file, illustrating the required environment variables for running the system locally. Actual secrets should never be committed or shared.

```
1 # Server
2 PORT=8080
3 NODE_ENV=development
4 LOG_LEVEL=info
5
6 # WebSocket Gateway
7 WS_MAX_PAYLOAD=1048576
8 WS_HEARTBEAT_INTERVAL_MS=18000
9
10 # Redis (Pub/Sub + cache)
11 REDIS_HOST=127.0.0.1
12 REDIS_PORT=6379
13
14 # Database
15 DB_ENGINE=postgres
16 DB_HOST=127.0.0.1
17 DB_PORT=5432
18 DB_USER=app_user
19 DB_NAME=messaging
20
21 # Auth (JWT)
22 JWT_ISSUER=example.com
23 JWT_AUDIENCE=messaging-clients
24 JWT_PUBLIC_KEY_PATH=./keys/jwt.pub
```

```

25
26 # Metrics
27 PROMETHEUS_PORT=9090

```

Listing A.1: Sample .env.example for local development (no secrets)

A.1.2 Gateway Configuration (YAML)

Here, a YAML configuration for the WebSocket gateway is provided, showing key parameters for connection limits, timeouts, and rate limiting.

```

1 gateway:
2   idleTimeoutMs: 30000
3   maxConcurrentConns: 80000
4   maxFrameSize: 262144
5   allowOrigins:
6     - https://app.example.com
7     - https://staging.example.com
8   rateLimit:
9     msgsPerSecond: 30
10    burst: 60
11    perIpConnLimit: 5

```

Listing A.2: WebSocket gateway configuration (sample)

A.2 Detailed API Specifications (REST & WebSocket)

This section documents the main API endpoints and WebSocket event schemas, supporting integration and client development.

A.2.1 WebSocket Events (Schemas)

This part describes the structure of WebSocket events exchanged between clients and the server, including message, presence, and typing events.

```

{
  "event": "msg.send",
  "data": {
    "channel_id": "c-82f5",
    "content": "Hello, team!",
    "client_ts": 1737112234
  },
  "trace_id": "8e4b0a7b-1c6a-4c5c-9f0e-1b2c3d4e5f6a"
}

```

Listing A.3: Client -> Server: msg.send

```
{
  "event": "msg.deliver",
  "data": {
    "message_id": "m-9ad12",
    "channel_id": "c-82f5",
    "server_ts": 1737112235,
    "sender_id": "u-42bd",
    "content": "Hello, team!"
  },
  "trace_id": "8e4b0a7b-1c6a-4c5c-9f0e-1b2c3d4e5f6a"
}
```

Listing A.4: Server -> Client: msg.deliver

```
{
  "event": "presence.update",
  "data": { "user_id": "u-42bd", "state": "online", "last_seen": 1737112237 }
}
```

Listing A.5: Presence and typing events (bidirectional)

A.2.2 REST Endpoints (Details)

Here, you will find sample REST API responses for authentication and message retrieval, useful for backend and frontend integration.

```
{
  "access_token": "eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9...",
  "expires_in": 3600,
  "token_type": "Bearer",
  "scope": "realtime:connect channels:read messages:write"
}
```

Listing A.6: POST /v1/auth/token (response)

```
{
  "page": 3,
  "page_size": 50,
  "items": [
    { "message_id": "m-9ad12", "sender_id": "u-42bd", "content": "Hello",
      "ts": 1737112201 },
    { "message_id": "m-9ad13", "sender_id": "u-51fa", "content": "Hi!",
      "ts": 1737112203 }
  ],
  "next_page": 4
}
```

Listing A.7: GET /v1/messages?channel=:id&page=:n (response)

A.2.3 Error Model and Rate Limits

This subsection explains the error response format and rate limiting strategies applied to API endpoints and WebSocket events.

```
{
  "error": {
    "code": "RATE_LIMIT_EXCEEDED",
    "message": "Too many requests",
    "request_id": "r-1f9d22",
    "hint": "Reduce frequency or back off and retry later."
  }
}
```

Listing A.8: Error envelope (all endpoints/events)

A.3 Additional Diagrams

This section includes full-size diagrams such as the entity relationship diagram, sequence diagrams, and deployment architecture to aid understanding of system structure and workflows.

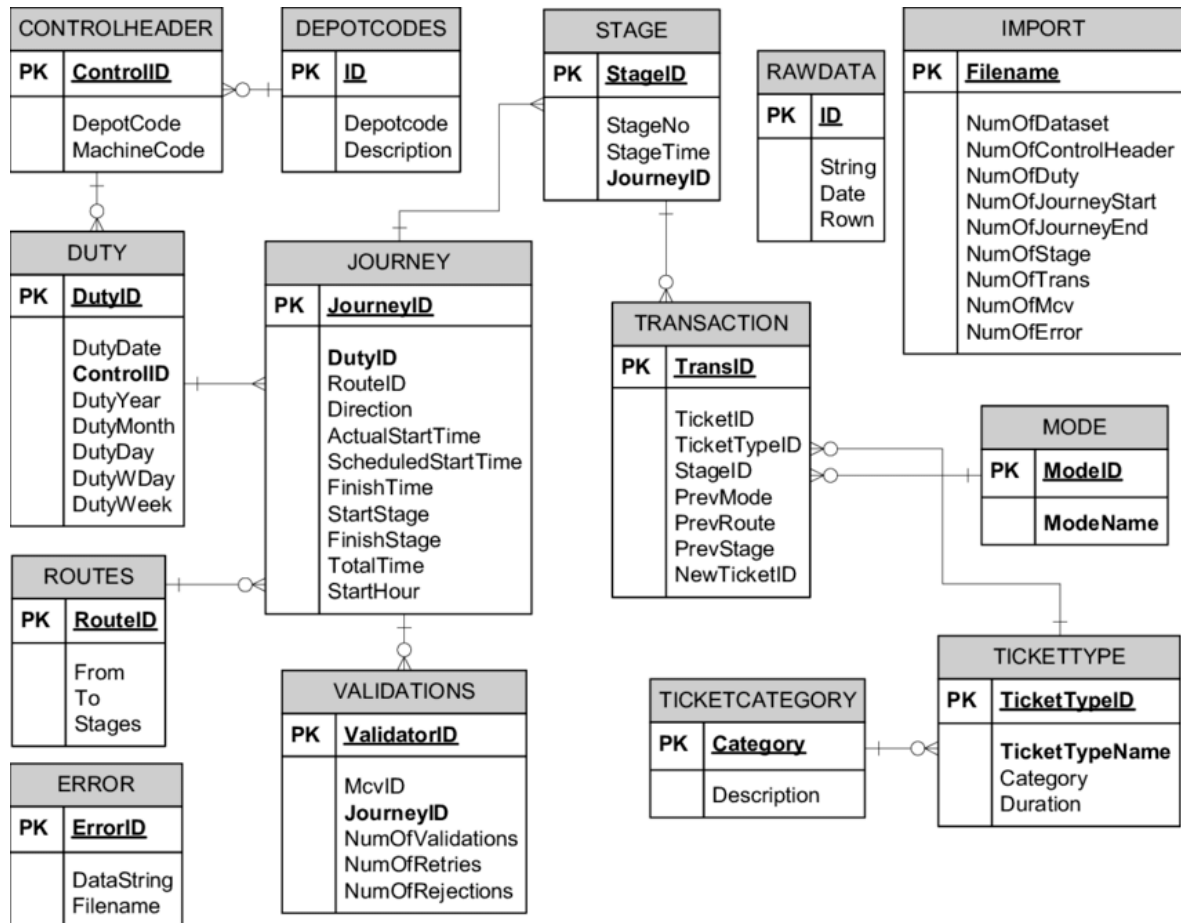


Figure A.1: Full-size Entity Relationship Diagram (ERD).

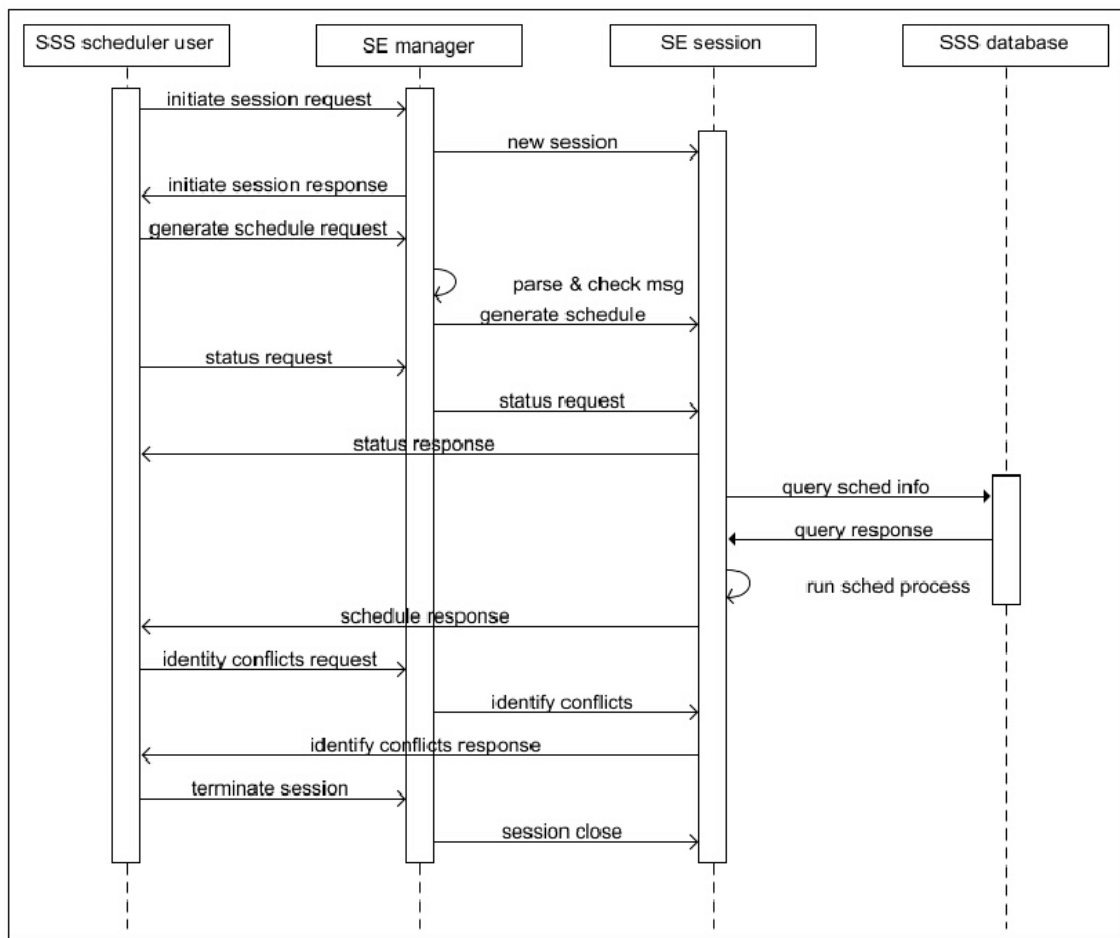


Figure A.2: Sequence diagram for message delivery pipeline.

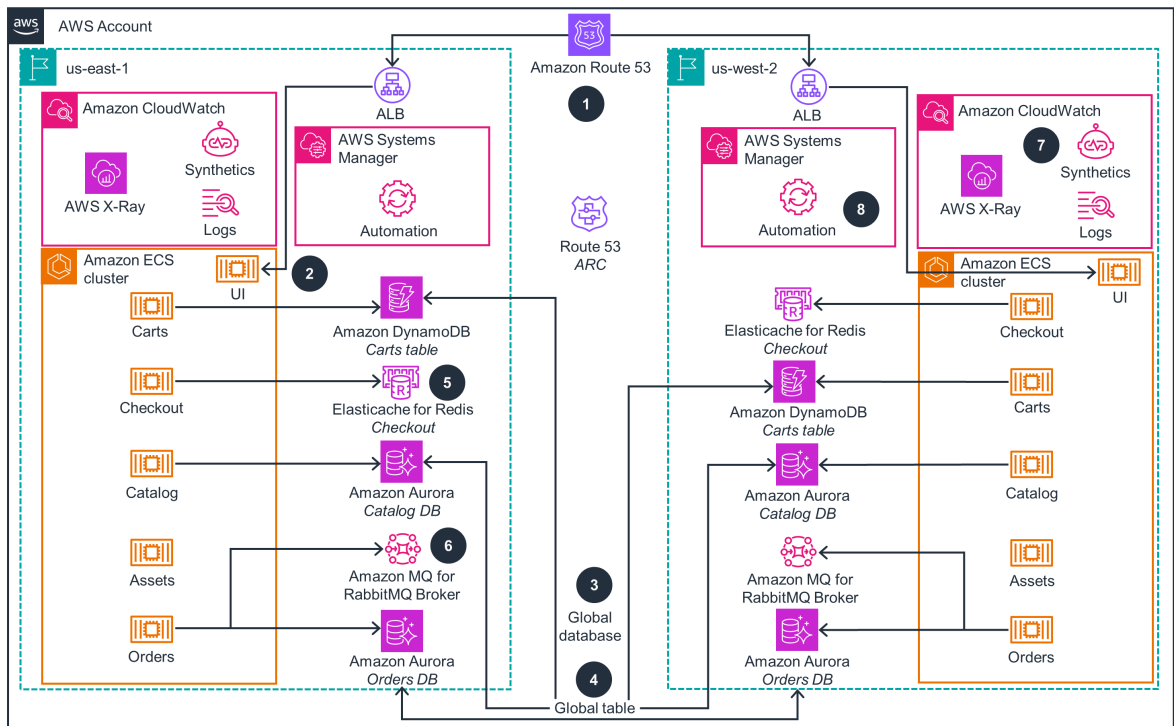


Figure A.3: AWS deployment architecture (detailed view).

A.4 Selected Code Snippets

Here, you will find practical code examples for key system components, including WebSocket handlers, Redis pub/sub integration, and Nginx proxy configuration.

A.4.1 WebSocket Message Handler (Node.js)

This snippet demonstrates how the WebSocket server processes incoming messages and delivers them to channel members.

```
ws.on('message', async (raw) => {
  try {
    const frame = JSON.parse(raw);
    if (frame.event === 'msg.send') {
      const saved = await messageService.persist(frame.data);
      await pubsub.publish(saved.channel_id, {
        event: 'msg.deliver',
        data: saved
      });
    }
  } catch (err) {
    logger.warn({ err }, 'invalid frame');
    ws.send(JSON.stringify({
      error: { code: 'BAD_REQUEST', message: 'Invalid JSON frame' }
    }));
  }
});
```

```

    }));
  }
});

```

Listing A.9: WebSocket message handler (simplified)

A.4.2 Redis Pub/Sub Integration

This example shows how Redis is used for pub/sub messaging, enabling efficient fan-out to multiple clients.

```

const sub = redis.duplicate();
await sub.connect();
await sub.subscribe(`chan:${channelId}`, (msg) => {
  const payload = JSON.parse(msg);
  fanoutService.broadcastToChannel(channelId, payload); // non-blocking
});

```

Listing A.10: Redis Pub/Sub fan-out (sketch)

A.4.3 Nginx Reverse Proxy for WebSocket

This configuration illustrates how Nginx is set up to proxy WebSocket connections securely and reliably.

```

1 server {
2   listen 443 ssl http2;
3   server_name ws.example.com;
4
5   location / {
6     proxy_pass http://gateway_pool;
7     proxy_set_header Upgrade $http_upgrade;
8     proxy_set_header Connection "Upgrade";
9     proxy_http_version 1.1;
10    proxy_read_timeout 60s;
11  }
12 }

```

Listing A.11: Nginx config snippet for WebSocket proxying

A.5 Load Testing Scripts

This section provides scripts and configuration files for load testing the messaging system, validating performance and scalability under stress.

A.5.1 k6 Script (WebSocket Burst)

This script uses k6 to simulate a burst of WebSocket connections and message traffic, testing system limits.


```

import ws from 'k6/ws';
import { check, sleep } from 'k6';

export const options = {
  scenarios: {
    burst: {
      executor: 'constant-vus',
      vus: 500,
      duration: '60s',
    }
  }
};

export default function () {
  const url = 'wss://ws.example.com';
  const params = { headers: { Authorization: `Bearer ${__ENV.TOKEN}` } };
  const res = ws.connect(url, params, function (socket) {
    socket.on('open', () => {
      for (let i = 0; i < 20; i++) {
        socket.send(JSON.stringify({ event: 'msg.send', data: { channel_id:
          'c-1', content: `hello ${i}` } }));
      }
    });
    socket.on('message', (msg) => { /* validate */ });
    socket.setTimeout(function () { socket.close(); }, 5000);
  });
  check(res, { 'status is 101': (r) => r && r.status === 101 });
  sleep(1);
}

```

Listing A.12: k6 WebSocket burst test (example)

A.5.2 Artillery (Channel Fan-out)

This Artillery configuration is designed to test channel fan-out performance by broadcasting messages to many recipients.

```

1 config:
2   target: "wss://ws.example.com"
3   phases:
4     - duration: 120
5       arrivalRate: 200
6   engines:
7     ws: {}
8   scenarios:
9     - engine: "ws"

```

```

10     flow:
11         - send:
12             event: "msg.send"
13             data:
14                 channel_id: "c-1"
15                 content: "broadcast"
16         - think: 1

```

Listing A.13: Artillery config for fan-out test (excerpt)

A.6 Evaluation Data Tables

This section contains tables summarizing latency, throughput, and resource usage metrics collected during system evaluation.

Table A.1: Latency distribution under load (single region, 3 nodes)

Scenario	p50 (ms)	p95 (ms)	p99 (ms)	Error %
1k conns, 2k msg/s	38	112	197	0.05
10k conns, 8k msg/s	45	138	265	0.12
30k conns, 15k msg/s	59	171	311	0.25

Table A.2: Throughput vs number of gateway nodes

Nodes	Msgs/s (avg)	CPU per node (%)	Redis CPU (%)
1	2,900	78	31
3	8,700	72	44
6	17,200	68	59

A.7 Deployment Scripts (Excerpts)

Here, you will find excerpts from deployment scripts, including Dockerfiles, docker-compose setups, and Terraform snippets for cloud infrastructure.

A.7.1 Dockerfile (Gateway)

This Dockerfile shows how the Node.js gateway service is built and configured for production deployment.

```

1 FROM node:20-alpine
2 WORKDIR /app
3 COPY package*.json ./
4 RUN npm ci --omit=dev
5 COPY . .
6 ENV NODE_ENV=production
7 EXPOSE 8080

```

```
8 CMD ["node", "dist/main.js"]
```

Listing A.14: Dockerfile (Node.js gateway, simplified)

A.7.2 docker-compose.yml (Local Dev)

This docker-compose file provides a local development environment with all required services, including Redis and PostgreSQL.

```
1 version: "3.9"
2 services:
3   redis:
4     image: redis:7-alpine
5     ports: ["6379:6379"]
6   db:
7     image: postgres:16-alpine
8     environment:
9       POSTGRES_DB: messaging
10      POSTGRES_USER: app_user
11      POSTGRES_PASSWORD: app_pass
12     ports: ["5432:5432"]
13   gateway:
14     build: .
15     environment:
16       - REDIS_HOST=redis
17       - DB_HOST=db
18     ports: ["8080:8080"]
19     depends_on: [redis, db]
```

Listing A.15: docker-compose (local development)

A.7.3 Terraform Snippet (ALB Listener for WSS)

This Terraform snippet demonstrates how to configure an AWS Application Load Balancer listener for secure WebSocket traffic.

```
1 resource "aws_lb_listener" "wss" {
2   load_balancer_arn = aws_lb.main.arn
3   port              = "443"
4   protocol          = "HTTPS"
5   ssl_policy        = "ELBSecurityPolicy-TLS-1-2-2017-01"
6   certificate_arn    = var.acm_cert_arn
7   default_action {
8     type            = "forward"
9     target_group_arn = aws_lb_target_group.gateway.arn
10  }
11 }
```

Listing A.16: Terraform (HCL) snippet for ALB listener (pseudo)

A.8 Experiment Reproducibility Checklist

This section lists the software versions and step-by-step instructions needed to reproduce the experiments and benchmarks described in the thesis.

A.8.1 Software Versions

This table details the exact software versions used in all experiments to ensure reproducibility and consistency.

Table A.3: Versions used in experiments

Component	Version
Node.js	20.x LTS
Redis	7.x
PostgreSQL	16.x
k6	0.50+
Artillery	2.x
Nginx	1.25+

A.8.2 Steps to Reproduce (Summary)

This summary outlines the main steps required to set up the infrastructure, deploy the system, and run performance tests.

1. Provision infrastructure (VPC, ALB, compute, ElastiCache, DB).
2. Deploy gateway container(s) and set environment variables (no secrets in images).
3. Seed database and create test channels/users.
4. Run k6/Artillery scenarios from a separate load injector VPC/subnet.
5. Collect metrics (CloudWatch, Prometheus, logs) and export CSV for analysis.

A.9 Additional Logs and Error Samples

This section provides sample logs and error codes to help with troubleshooting and understanding system behavior during failures.

A.9.1 Structured Log (Masked)

This log example shows the format and key fields of structured logs generated by the gateway service.

```
{
  "ts": "2026-01-17T14:20:01.120Z",
  "level": "info",
  "msg": "delivered",
  "channel": "c-82f5",
```

```

    "message_id": "m-9ad12",
    "latency_ms": 42,
    "conn_id": "k3r2",
    "trace_id": "8e4b0a7b-1c6a-4c5c-9f0e-1b2c3d4e5f6a"
  }

```

Listing A.17: Gateway structured log (JSON)

A.9.2 WebSocket Close Codes (Excerpt)

This table lists common WebSocket close codes and their meanings, useful for diagnosing connection issues.

Table A.4: Common WebSocket close codes (excerpt)

Code	Name	Meaning
1000	Normal Closure	Regular closure without error
1006	Abnormal Closure	No close frame; network break or peer lost
1008	Policy Violation	Server closed due to policy (e.g., ACL)
1011	Internal Error	Unexpected server condition

A.10 Glossary of Technical Terms

This glossary defines important technical terms used throughout the appendix and thesis, supporting reader understanding.

- **Backpressure:** Mechanisms to prevent producers from overwhelming consumers by applying flow control.
- **Fan-out:** Broadcasting a message to multiple recipients (e.g., all members of a channel).
- **Presence:** Representation of a user's online/offline/away state and last-seen timestamp.
- **Sharding:** Partitioning data or load across multiple nodes to scale horizontally.
- **Sticky Session:** Routing a client to the same backend node across requests or connections.
- **Trace ID:** Unique identifier propagated across services for distributed tracing.